

SUBMITTAL RECORD

Norcross, GA — Lidl #1517 | Job 15170000GR
4990 Jimmy Carter Blvd, Norcross, GA 30093



Submittal Information

PSI Log #	SUB-010 (Rev 0)	Procore #	1-1.0 — Mast Arm Pole Submittals
Fieldwire #	PSI-0005 — Mast Arm Pole Submittals	Type	Product Data — structural calculations (Procore: Cutsheet)
Spec Section	None assigned — Procore "1 - Default Section"	Status	Pending — no response. OVERDUE (Fieldwire response due 09/25/2026)
Subcontractor	R.J. Haynie & Associates; manufacturer Valmont Industries (Valley, NE) — Order 672263-P1, prepared by Skyler Ellefson 8/18/2026	Submitted By	Tucker Teague (Place Services Inc.) — PSI stamp No Exception Taken 9/18/2026
Scope / Location	Off-site traffic signal structures: Gwinnett County TS-082, Jimmy Carter Blvd @ Shelby Dr	Ball in Court	Christopher Manes (CHA Consulting, Inc.)
Issue Date	Sep 18, 2026	Final Due Date	None set in Procore
Procore Received	Sep 25, 2026	Related	SUB-009 (Procore 1-2.0) — Valmont shop drawings GA672263P1 for the same poles
Distribution	Christopher Manes, JaWanna English-Neal, Jeff Belyea, Todd Seldomridge (CHA Consulting); Jens Byer (Lidl US); Bobby McTeague, Luis Marcelino (Place Services Inc.)		

Package Contents (per Valmont Table of Contents)

Calc pp. 1–47	Pole B — P22' with M65' & M55' mast arms, AASHTO 2015, 115 mph ultimate wind (700-yr MRI), fatigue Cat. 2, ice included. 16.00"/12.92" x 22' x 0.3125" A572 shaft
Calc pp. 48–94	Pole C — P30' with M55' & M55' mast arms, AASHTO 2015, 115 mph
Each pole	Strength I / Extreme I / Service I / Ice / natural wind gust load cases; anchor bolt analysis; "Reactions Applied To Foundation" (calc pp. 6 and 53)

Submittal Workflow / Revision History

Rev	Name	Sent	Due	Returned	Response	Attachment
0	Christopher Manes (CHA Consulting)	Sep 18, 2026	Sep 25, 2026*	—	None — OVERDUE 6 days	#1517 Lidl Norcross, GA - Mast Arm Pole Submittals.pdf

* No Final Due Date in Procore; due date taken from Fieldwire PSI-0005 "Response due 09/25/2026".

Reviewer Comments / Notes

No engineer review stamp or markups as of Oct 1, 2026.

Calcs provide foundation reactions only — no foundation (drilled shaft) design is included. The 3'-0" x 14'-6" (8) #10 foundation note on SUB-009 Dwg #2 is not supported by this package; confirm foundation design source (signal plans / county standard).

GA PE seal (Nizam A Qassem, PE039283) is applied on the cover as an image stamp dated 8/31/2026; unlike the SUB-009 drawings, no digital signature is present. Confirm signed/sealed calcs if the county requires them.

Enclosed (merged behind this sheet, unaltered)

Full Procore export as received Oct 1, 2026 — 98 pages: p1 Procore cover; p2 Fieldwire PSI-0005 cover; p3 Valmont proposal cover with PE seal and PSI stamp; p4 table of contents; p5–98 calculations (calc pp. 1–94).

Prepared by Luis Marcelino, Place Services, Inc. — Oct 1, 2026



Lidl US, LLC
3500 S. Clark St.
Arlington , Virginia 22202
P: (571) 398-5435

Project: 1517 Norcross GA - Jimmy Carter Blvd
4990 Jimmy Carter Blvd
Norcross, Georgia 30093

Submittal #1-1.0 - Mast Arm Pole Submittals 1 - Default Section

Revision	0	Submittal Manager	Christopher Manes (21327198 CHA Consulting, INC.)
Status	Open	Date Created	Sep 18, 2026
Issue Date	Sep 18, 2026	Spec Section	1 - Default Section
Responsible Contractor	Place Services Inc.	Received From	Tucker Teague (Place Services Inc.)
Received Date	Sep 25, 2026	Submit By	Sep 18, 2026
Final Due Date		Lead Time	
		Cost Code	
Location		Type	Cutsheet
Submittal Package			
Approvers			
Ball in Court	Christopher Manes (21327198 CHA Consulting, INC.)		
Distribution	Bobby McTeague (Place Services Inc.), Christopher Manes (21327198 CHA Consulting, INC.), JaWanna English-Neal (21327198 CHA Consulting, INC.), Jeff Belyea (21327198 CHA Consulting, INC.), Jens Byer (28101155 Lidl US, LLC), Luis Marcelino (Place Services Inc.), Todd Seldomridge (21327198 CHA Consulting, INC.)		
Description	Place Services Inc. - No Exceptions Taken		

Submittal Workflow

Name	Sent Date	Due Date	Returned Date	Response	Attachments
General Information Attachments					#1517 Lidl Norcross, GA - Mast Arm Pole Submittals.pdf

Lidl 1517 Norcross, GA - Ground Up

4990 Jimmy Carter Blvd. Norcorss, GA 30093

Report generated by Tucker Teague (Place Services, Inc.)



Submittal #PSI-0005 - Mast Arm Pole Submittals

Creator	Tucker Teague (Place Services, Inc.)	Submission due	-
Assignee	Tucker Teague (Place Services, Inc.)	Response due	09/25/2026
Reviewers	-	Closed on	-
Status	Pending approval	Reference number	-
Type	Product data	Specification	-
Description	-		

Approval (rev. 0)

Responded by	-	Responded on	-
Response	-		
Notes	-		
Attachments	-		

Submission (rev. 0)

Submitted by	Tucker Teague (Place Services, Inc.)	Submitted on	09/18/2026
Notes	Place Services Inc. - No Exceptions Taken		
Attachments	PDF attachments are included in this export. ↓ Mast Arm Pole Submittals.pdf (Page 2)		



Valmont Industries, Inc.
West Highway 275
P.O. Box 358
Valley, Nebraska 68064-0358 USA
(402) 359-2201

A Light & Traffic Structure Proposal
for
Jimmy Carter Blvd at Shelby Dr
Gwinnett County, Georgia

Valmont Order No.: 672263-P1



Prepared By:
Skyler Ellefson
August 18, 2026

**SHOP DRAWING /
SUBMITTAL REVIEW**

- ☒ NO EXCEPTION TAKEN
☐ REVISE AND RESUBMIT
☐ MAKE CORRECTIONS NOTED
☐ REJECTED

SUBMITTAL WAS REVIEWED FOR DESIGN CONFORMITY AND GENERAL CONFORMANCE TO CONTRACT DOCUMENTS ONLY. THE CONTRACTOR IS RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AT JOBSITE FOR TOLERANCE, CLEARANCE, QUANTITIES, FABRICATION PROCESSES AND TECHNIQUES OF CONSTRUCTION, COORDINATION OF HIS WORK WITH OTHER TRADES AND FULL COMPLIANCE WITH CONTRACT DOCUMENTS.

By: Tucker Teague
Date: 9/18/2026

Place Services, Inc
Canton, GA 30115

Proprietary Information

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ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
IN ACCORDANCE WITH AASHTO-2015 RQMTS. (FINAL DEFLECTED POSITION)

BY: SE70

08/18/2026

VERSION: 29.0.17.35

SUBJECT: JIMMY CARTER BLVD-POLE B-P22' M65'x55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P22M6555A15115

Design Criteria

Design Code	AASHTO-2015	Fatigue Category	2
Ultimate Wind Speed (mph)	115.0	Truck Gust	No
Mean Recurrence Interval	700	Galloping	No
Service Level Wind Speed (mph)	76.0	Natural Wind Gust	Yes
AASHTO Ice Included ?	Yes		

Design Summary - Pole

Height (ft)	Shaft Weight (lb)	Ground Line Diameter (in)	Top Dia. (in)
22.0000	1039	16.00	12.920

Section Characteristics

	Section - 1
Shape	Round
Top Dia. (in)	12.920
Base Diameter (in)	16.000
Thickness (in)	0.31250
Length (ft)	22.00
Shaft Weight (lb)	1039
Assembly Weight (lb)	1573
Taper (in/ft)	0.14000
Yield Strength (ksi)	55.00
Material	S220 - A572

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Base Plate

Shape	Square
Material	S70 - A36
Width (in)	23.000
Thickness (in)	2.50000
Yield Strength (ksi)	36.00
Base Weld Type	SOCKET
Weight (lb)	207

Anchor Bolts

Material	S100 - F1554
Bolt diameter (in)	2.25
Bolt circle diameter (in)	22.50
Quantity	4
Yield Strength (ksi)	55.00
Tensile strength (ksi)	75

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Design Summary - Arms

Signal and Sign	Arm 1	Arm 2
Shape	Round	Round
Span Length (ft)	65.0000	55.0000
Taper (in/ft)	0.14000	0.14000
Attachment Height (ft)	20.00	20.00
Orientation (deg)	180.00	270.00
Slope at Base (deg)	0.00	0.00
Centroid Location		
Horizontal (ft)	28.0454	24.3929
Above Attachment (ft)	0.0000	0.0000
Unbent Length (ft)	65.0000	55.0000
Material-Base	S220 - 55 ksi	S220 - 55 ksi
Weight (lb)	1741	1399
Base Section		
Base O.D. (in)	15.00	14.50
Thickness (in)	0.31250	0.25000
Length (ft)	19.5416	22.9880
Yield Strength (ksi)	55.00	55.00
Material	S220	S220
Joint Type	Slip Joint	Slip Joint
Overlap Length (ft)	2.6945	2.5695
Outer Section		
Base O.D. (in)	13.00	12.00
Thickness (in)	0.17930	0.17930
Length (ft)	48.1529	34.5814
Yield Strength (ksi)	55.00	55.00
Material	S105	S105

Base Weld Type = Socket

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SUBJECT: JIMMY CARTER BLVD-POLE B-P22' M65'x55' AASHTO 2015 115MPH
Folder: GAGWIN
Design Summary - Arms

VERSION: 29.0.17.35

File: P22M6555A15115

Simplex Dimensions

	Arm 1	Arm 2
Connection Bolt Data		
Number of bolts	4	4
Bolt diameter (in)	1.25	1.25
ASTM Specification	A325	A325
Horizontal Spacing (in)	19.00	19.00
Vertical Spacing (in)	19.00	19.00
Attachment Plate Data		
Horizontal Width (in)	24.00	24.00
Vertical Width (in)	24.00	24.00
Mast Arm Bracket Thickness (in)	3.00	3.00
Arm Plate Bracket Weight (lb)	242	247
Pole Plate Bracket Thickness (in)	2.00	2.00
Pole Plate Bracket Weight (lb)	161	165
Yield Strength (ksi)	36.00	36.00
Vertical Gusset Thickness (in)	0.5000	0.5000
Horizontal Gusset Thickness (in)	0.5000	0.5000

Attachment Type

Arm 1:	SIMPLEX - THRU BOLTS,	Base Weld Type = Socket
Arm 2:	SIMPLEX - THRU BOLTS,	Base Weld Type = Socket

** These heights are above bottom of base plate or transformer base.

Elliptical cross section; first diameter is horizontal.

** Arm orientations are angles from +X axis in X-Y plane.

X and Y axes are perpendicular/parallel to sides of pole base plate. See *** below.

*** If arm is attached with a clamp, height and orientation must not be changed from values shown above without consulting Valmont.

Nice to have:

**** Assembly weight includes unfinished shaft + flange + simplex plate.

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Description of Sign Loading

Position of Signal or Sign	Mounting Height ** (ft)	Centroid Height ** (ft)	Distance To Centroid From Pole (ft)	Sign Weight (lb)	Sign Width (ft)	Sign Height (ft)	Sign Depth (ft)	Sign Cd
Mast Arm 1	20.0000	20.0000	43.0000	30	10.0000	1.500	0.000	1.23
Mast Arm 2	20.0000	20.0000	26.0000	20	6.5000	1.500	0.000	1.20
Mast Arm 2	20.0000	20.0000	38.0000	15	2.5000	3.000	0.000	1.19

Description of Luminaire/Signal Loading

Position of Signal	Mounting Height ** (ft)	Centroid Height ** (ft)	Distance To Centroid From Pole (ft)	Weight (lb)	Vertical Plane Area (ft2)	Horizontal Plane Area (ft2)
Arm 1	20.0000	20.0000	55.0000	50	8.67	1.80
Arm 1	20.0000	20.0000	63.0000	50	8.67	1.80
Arm 2	20.0000	20.0000	20.0000	50	8.67	1.80
Arm 2	20.0000	20.0000	32.0000	50	8.67	1.80
Arm 2	20.0000	20.0000	44.0000	62	11.20	1.80

THE VALUES SHOWN IN THIS TABLE MUST NOT BE EXCEEDED
WITHOUT CONSULTING VALMONT.

ANY SIZES OR OTHER DIMENSIONS NOT PROVIDED BY THE
SPECIFYING AGENCY HAVE
BEEN ESTIMATED BY VALMONT.

** THESE HEIGHTS ARE ABOVE BOTTOM OF BASE PLATE OR
TRANSFORMER BASE.

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RESULTS SUMMARY - Pole

Maximum Combined Force Interaction In Each Major Component		Maximum Reactions Applied To Foundation	
Strength I		Bending Moment	
Pole CFI (At 20.00 (ft))	0.39	Torsion	146925 ft-lb
Signal and Sign Arm 1 CFI	0.26	Shear Force	80953 ft-lb
Signal and Sign Arm 2 CFI	0.25	Axial Force	3259 lb
Base Plate CFI	0.17		7400 lb
Anchor Bolts CFI	0.27	Ice	
S/S Arm 1 Att. Bolts CFI	0.05	Pole CFI (At 20.00 (ft))	0.57
S/S Arm 1 Att. Plate CFI	0.09	Signal and Sign Arm 1 CFI	0.48
S/S Pole 1 Att. Plate CFI	0.19	Signal and Sign Arm 2 CFI	0.41
S/S Arm 2 Att. Bolts CFI	0.03	Base Plate CFI	0.31
S/S Arm 2 Att. Plate CFI	0.07	Anchor Bolts CFI	0.48
S/S Pole 2 Att. Plate CFI	0.15	S/S Arm 1 Att. Bolts CFI	0.21
Deflection % (At 22.00 (ft))	1.158 %	S/S Arm 1 Att. Plate CFI	0.18
Deflection (At 22.00 (ft))	3.06 in	S/S Pole 1 Att. Plate CFI	0.40
Rotation (At 22.00 (ft))	1.32 deg	S/S Arm 2 Att. Bolts CFI	0.14
Extreme I		S/S Arm 2 Att. Plate CFI	0.14
Pole CFI (At 20.00 (ft))	0.50	S/S Pole 2 Att. Plate CFI	0.33
Signal and Sign Arm 1 CFI	0.51	Deflection % (At 22.00	1.861
Signal and Sign Arm 2 CFI	0.42	(ft))	%
Base Plate CFI	0.27	Deflection (At 22.00 (ft))	4.91 in
Anchor Bolts CFI	0.42	Rotation (At 22.00 (ft))	2.07 deg
S/S Arm 1 Att. Bolts CFI	0.23	Fatigue II	
S/S Arm 1 Att. Plate CFI	0.18	Pole (At 0.00 (ft))	0.56
S/S Pole 1 Att. Plate CFI	0.41	Hand Hole / Opening	0.55
S/S Arm 2 Att. Bolts CFI	0.15	Signal and Sign Arm 1	
S/S Arm 2 Att. Plate CFI	0.15	Arm Tube	0.80
S/S Pole 2 Att. Plate CFI	0.34	Simplex Bolts	0.74
Deflection % (At 22.00 (ft))	1.382 %	Signal and Sign Arm 2	
Deflection (At 22.00 (ft))	3.65 in	Arm Tube	0.87
Rotation (At 22.00 (ft))	1.46 deg	Simplex Bolts	0.60
Service I		Anchor Bolts	0.26
Pole CFI (At 0.00 (ft))	0.32		
Signal and Sign Arm 1 CFI	0.30		
Signal and Sign Arm 2 CFI	0.26		
Base Plate CFI	0.19		
Anchor Bolts CFI	0.30		
S/S Arm 1 Att. Bolts CFI	0.10		
S/S Arm 1 Att. Plate CFI	0.12		
S/S Pole 1 Att. Plate CFI	0.27		
S/S Arm 2 Att. Bolts CFI	0.06		
S/S Arm 2 Att. Plate CFI	0.10		
S/S Pole 2 Att. Plate CFI	0.22		
Deflection % (At 22.00 (ft))	1.140 %		
Deflection (At 22.00 (ft))	3.01 in		
Rotation (At 22.00 (ft))	1.24 deg		

SHAFT PROPERTIES: NUMER OF SIDES = Round
DIAMETER (D) = 16.000 IN
THICKNESS (T) = 0.31250 IN
YIELD STRENGTH (KSI) = 55
MOM. OF INERTIA (IX) = 473.138 IN^4
SECTION MODULUS (SX) = 59.142 IN^3

MN: ALLOWABLE SHAFT MOMENT

MB = (Fb * Sx) / 12

= (55000 * 59.142 / 12 = 271069

MB = 271069 FT-LB

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
IN ACCORDANCE WITH AASHTO-2015 RQMTS. (FINAL DEFLECTED POSITION)

BY: SE70

08/18/2026

VERSION: 29.0.17.35

SUBJECT: JIMMY CARTER BLVD-POLE B-P22' M65'x55' AASHTO 2015 115MPH

Folder: GAGWIN
Pole Properties

File: P22M6555A15115

Height (ft)	Diameter (in)	Wall Thk. (in)	Roundness Ratio (%)	D/t	B/T	Moments of Inertia (in4)	Plastic Section Modulus (in3)	Area (in2)	Radius of Gyration (in)
22.0000	12.92	0.31250	100.0	41.34	0.00	245.61	49.63	12.37	4.46
20.0000	13.20	0.31250	100.0	42.24	0.00	262.33	51.86	12.65	4.55
18.5000	13.41	0.31250	100.0	42.91	0.00	275.37	53.56	12.85	4.63
17.0000	13.62	0.31250	100.0	43.58	0.00	288.82	55.29	13.06	4.70
15.3333	13.85	0.31250	100.0	44.33	0.00	304.28	57.25	13.29	4.79
13.6667	14.09	0.31250	100.0	45.08	0.00	320.29	59.24	13.52	4.87
12.0000	14.32	0.31250	100.0	45.82	0.00	336.84	61.26	13.75	4.95
10.3333	14.55	0.31250	100.0	46.57	0.00	353.95	63.32	13.98	5.03
8.6667	14.79	0.31250	100.0	47.32	0.00	371.63	65.41	14.21	5.11
7.0000	15.02	0.31250	100.0	48.06	0.00	389.90	67.54	14.43	5.20
5.3333	15.25	0.31250	100.0	48.81	0.00	408.75	69.70	14.66	5.28
3.6667	15.49	0.31250	100.0	49.56	0.00	428.20	71.89	14.89	5.36
2.0000	15.72	0.31250	100.0	50.30	0.00	448.25	74.12	15.12	5.44
0.0000	16.00	0.31250	100.0	51.20	0.00	473.14	76.84	15.40	5.54

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Arm: Forces and Moments (Strength I)

Arm Type	Arm No.	Analysis Location	Forces (lb)			Moment (ft-lb)		
			Axial	Fy	Fz	Torsion	My	Mz
SIGNAL	1	BASE	49.11	33.93	2,404.17	0.00	62,559.48	882.77
SIGNAL	1	SPLICE-I	45.39	20.22	1,432.63	0.00	30,460.29	429.82
SIGNAL	1	SPLICE-O	39.66	17.07	1,209.72	0.00	26,900.37	379.59
SIGNAL	2	BASE	33.67	37.41	2,051.80	0.00	48,306.26	880.86
SIGNAL	2	SPLICE-I	32.25	19.88	1,090.38	0.00	15,938.39	290.63
SIGNAL	2	SPLICE-O	28.33	16.74	917.87	0.00	13,358.28	243.59

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Arm: Resistances (Strength I)

Analysis Location			Comb. Force Inter.	Applied Forces			Factored Resistance Forces		
Arm Type	Arm No.	Site		Axial (lb)	Shear (lb)	Bend. (ft-lb)	Axial $\phi=0.9$ (lb)	Shear $\phi=0.9$ (lb)	Bend. $\phi=0.9$ (ft-lb)
SIGNAL	1	BASE	0.24	49	2,404	62,566	713,545	214,064	262,762
SIGNAL	1	SPLICE-I	0.16	45	1,433	30,463	598,958	179,687	191,565
SIGNAL	1	SPLICE-O	0.26	40	1,210	26,903	346,855	104,057	101,816
SIGNAL	2	BASE	0.25	34	2,052	48,314	553,834	166,150	191,534
SIGNAL	2	SPLICE-I	0.13	32	1,091	15,941	442,731	132,819	127,171
SIGNAL	2	SPLICE-O	0.15	28	918	13,360	319,468	95,840	87,354

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Arm Connection Analysis (Strength I)

Analysis of Signal/Sign Arm Simplex Bolts

Mast Arm	Max Bolt CFI	Applied Forces (kip)		Factored Resistance (kip)	
		Tension	Shear	Tension	Shear
1	0.05	20.05	0.60	89.54	44.77
2	0.03	15.54	0.51	89.54	44.77

Analysis of Signal/Sign Arm Simplex Plates

Member Type	Max CSR	Applied Moment (ft-lb)	Factored Resistance (ft-lb)	Angle of Failure Line (deg)	Length of Bend Line (in)
1	0.09	9,868.93	114,733.70	45	18.89
2	0.07	7,975.59	117,779.92	45	19.39

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Pole: Forces and Moments (Strength I)

Section Height* (ft)	Shear (lb)		Axial (lb)	Moment (ft-lb)		Torsion (ft-lb)	Wind Direct** (deg)
	Fx	Fy	Fz	Mx	My	Mz	
20.00	-79.20	-61.29	4,565.06	49,436.32	-63,883.82	0.00	0.00
0.00	-4.03	-3.12	5,785.92	50,066.31	-64,697.91	0.00	0.00

* These heights are above the pole base plate.

** These are directions toward which the wind is flowing.
They are angles from the +X axis in the X-Y plane.

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
IN ACCORDANCE WITH AASHTO-2015 RQMTS. (FINAL DEFLECTED POSITION)

BY: SE70

08/18/2026

VERSION: 29.0.17.35

SUBJECT: JIMMY CARTER BLVD-POLE B-P22' M65'&55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P22M6555A15115

Pole: Resistances (Strength I)

Section Height* (ft)	Comb. Force Inter.	Applied Force				Factored Resistance			
		Axial (lb)	Shear (lb)	Bend. (ft-lb)	Torsion (ft-lb)	Axial $\phi=0.9$ (lb)	Shear $\phi=0.9$ (lb)	Bend. $\phi=0.9$ (ft-lb)	Torsion $\phi=0.95$ (ft-lb)
20.00	0.39	4,565.06	100.14	80,778.04	0.00	619,959.19	187,828.86	207,429.51	212,884.15
0.00	0.28	5,785.92	5.10	81,807.43	0.00	340,928.99	228,638.31	296,279.46	315,437.59

* These heights are above the pole base plate.

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Baseplate Analysis (Strength I) - Pole1 - Pole

Combined Force Interaction	0.17
Critical Wind Direction *	0.00 deg
Alignment of Bend Line	135.00 deg
Width of Bending Section	16.527 (in)
Failure Line Start Coordinate (in)	(-11.500, 0.186)
Failure Line End Coordinate in	(0.186, -11.500)
Applied Bending Moment	12,113.50 ft-lb
Factored Bending Resistance	69,722.91 ft-lb
Plate Controlling Bolt Forces	

Bolt Number	Axial force (lb)	Moment Arm (in)	Bending moment (ft-lb)
3	-44727	3.25	-12114

Anchor Bolts Analysis (Strength I) - Pole1 - Pole

Critical Wind Direct.* (deg)	Comb. Force Inter.	Applied Stress (psi)		Factored Resistance (psi)	
		Axial	Shear	$\phi F'_{nt}$	ϕF_v
0.00	0.27	11,248.97	0.00	42,187.50	22,500.00

* Per AISC Design Guide 1

* These are directions toward which the wind is flowing

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Pole Deflection Information: (Strength I)

Critical Wind Direction: 0.00

Height (ft)	X-Defl. (in)	Y-Defl. (in)	Deflection Angle - X (deg)	Deflection Angle - Y (deg)
22.0000	-2.42	-1.87	-1.04	0.81
20.0000	-1.98	-1.53	-1.04	0.81
18.5000	-1.67	-1.29	-0.94	0.73
17.0000	-1.39	-1.07	-0.85	0.65
15.3333	-1.11	-0.86	-0.74	0.58
13.6667	-0.87	-0.67	-0.65	0.50
12.0000	-0.66	-0.51	-0.55	0.43
10.3333	-0.48	-0.37	-0.46	0.36
8.6667	-0.33	-0.26	-0.38	0.29
7.0000	-0.21	-0.17	-0.30	0.23
5.3333	-0.12	-0.09	-0.22	0.17
3.6667	-0.06	-0.04	-0.15	0.12
2.0000	-0.02	-0.01	-0.08	0.06
0.0000	0.00	0.00	0.00	0.00

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Folder: GAGWIN

File: P22M6555A15115

EXTREME I LIMIT STATE

Wind Velocity	115.0 mph
Dead Component Load Factor	1.10
Wind Load Factor	1.00
Gust Factor	1.14

Mast Arm 1: Wind and Weight Force Data (Extreme I)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft2)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)	Notes
ATTCHMT. 1	20.0000	43.0000	15.00	1.23	0.90	0.85	29.44	543	
ATTCHMT. 2	20.0000	55.0000	8.67	1.20	0.90	0.85	29.44	306	
ATTCHMT. 3	20.0000	63.0000	8.67	1.20	0.90	0.85	29.44	306	

Mast Arm 2: Wind and Weight Force Data (Extreme I)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft2)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)	Notes
ATTCHMT. 4	20.0000	20.0000	8.67	1.20	0.90	0.85	29.44	306	
ATTCHMT. 5	20.0000	26.0000	9.75	1.20	0.90	0.85	29.44	344	
ATTCHMT. 6	20.0000	32.0000	8.67	1.20	0.90	0.85	29.44	306	
ATTCHMT. 7	20.0000	38.0000	7.50	1.19	0.90	0.85	29.44	263	
ATTCHMT. 8	20.0000	44.0000	11.20	1.20	0.90	0.85	29.44	396	

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Arm: Forces and Moments (Extreme I)

Arm Type	Arm No.	Analysis Location	Forces (lb)			Moment (ft-lb)		
			Axial	Fy	Fz	Torsion	My	Mz
SIGNAL	1	BASE	98.34	1,952.32	2,076.79	4.66	53,363.53	81,532.46
SIGNAL	1	SPLICE-I	113.23	1,640.58	1,227.31	0.63	25,730.04	50,968.45
SIGNAL	1	SPLICE-O	111.82	1,636.58	1,031.09	0.02	22,687.33	46,553.08
SIGNAL	2	BASE	88.77	2,309.53	1,751.97	2.29	40,880.35	69,079.88
SIGNAL	2	SPLICE-I	102.70	1,655.91	920.30	0.05	13,421.24	25,328.75
SIGNAL	2	SPLICE-O	102.94	1,652.25	768.41	0.14	11,251.57	21,078.39

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Arm: Resistances (Extreme I)

Analysis Location			Comb. Force Inter.	Applied Forces			Factored Resistance Forces		
Arm Type	Arm No.	Site		Axial (lb)	Shear (lb)	Bend. (ft-lb)	Axial $\phi=0.9$ (lb)	Shear $\phi=0.9$ (lb)	Bend. $\phi=0.9$ (ft-lb)
SIGNAL	1	BASE	0.37	98	2,850	97,443	713,545	214,064	262,762
SIGNAL	1	SPLICE-I	0.30	113	2,049	57,095	598,958	179,687	191,565
SIGNAL	1	SPLICE-O	0.51	112	1,934	51,787	346,855	104,057	101,816
SIGNAL	2	BASE	0.42	89	2,899	80,270	553,834	166,150	191,534
SIGNAL	2	SPLICE-I	0.23	103	1,894	28,665	442,731	132,819	127,171
SIGNAL	2	SPLICE-O	0.27	103	1,822	23,893	319,468	95,840	87,354

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Arm Connection Analysis (Extreme I)

Analysis of Signal/Sign Arm Simplex Bolts

Mast Arm	Max Bolt CFI	Applied Forces (kip)		Factored Resistance (kip)	
		Tension	Shear	Tension	Shear
1	0.23	42.62	0.71	89.54	44.77
2	0.15	34.75	0.72	89.54	44.77

Analysis of Signal/Sign Arm Simplex Plates

Member Type	Max CSR	Applied Moment (ft-lb)	Factored Resistance (ft-lb)	Angle of Failure Line (deg)	Length of Bend Line (in)
1	0.18	20,983.35	114,733.70	45	18.89
2	0.15	17,831.56	117,779.92	45	19.39

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Pole: Wind and Weight Force Data (Extreme I)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft2)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)
ATTCHMT. 1	20.0000	43.0000	15.00	1.23	0.90	0.85	29.44	543
ATTCHMT. 2	20.0000	55.0000	8.67	1.20	0.90	0.85	29.44	306
ATTCHMT. 3	20.0000	63.0000	8.67	1.20	0.90	0.85	29.44	306
ATTCHMT. 4	20.0000	20.0000	8.67	1.20	0.90	0.85	29.44	306
ATTCHMT. 5	20.0000	26.0000	0.00	1.20	0.90	0.85	29.44	0
ATTCHMT. 6	20.0000	32.0000	8.67	1.20	0.90	0.85	29.44	306
ATTCHMT. 7	20.0000	38.0000	0.00	1.19	0.90	0.85	29.44	0
ATTCHMT. 8	20.0000	44.0000	11.20	1.20	0.90	0.85	29.44	396
22.0000	20.9964	0.0000	2.18	0.45	0.91	0.85	29.74	29
20.0000	19.2480	0.0000	1.66	0.45	0.89	0.85	29.20	22
18.5000	17.7481	0.0000	1.69	0.45	0.88	0.85	28.71	22
17.0000	16.1643	0.0000	1.91	0.45	0.86	0.85	28.15	24
15.3333	14.4977	0.0000	1.94	0.45	0.86	0.85	28.09	25
13.6667	12.8311	0.0000	1.97	0.45	0.86	0.85	28.09	25
12.0000	11.1644	0.0000	2.01	0.45	0.86	0.85	28.09	25
10.3333	9.4978	0.0000	2.04	0.45	0.86	0.85	28.09	26
8.6667	7.8312	0.0000	2.07	0.45	0.86	0.85	28.09	26
7.0000	6.1645	0.0000	2.10	0.45	0.86	0.85	28.09	27
5.3333	4.4979	0.0000	2.13	0.45	0.86	0.85	28.09	27
3.6667	2.8313	0.0000	2.17	0.45	0.86	0.85	28.09	27
2.0000	0.9971	0.0000	2.64	0.45	0.86	0.85	28.09	33

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Folder: GAGWIN

File: P22M6555A15115

Pole: Forces and Moments (Extreme I)

Section Height* (ft)	Shear (lb)		Axial (lb)	Moment (ft-lb)		Torsion (ft-lb)	Wind Direct** (deg)
	Fx	Fy	Fz	Mx	My	Mz	
20.00	-141.61	-3,024.73	3,959.84	42,505.04	-54,498.30	80,951.29	270.00
0.00	-6.60	-3,263.84	5,091.77	105,696.47	-55,525.55	80,951.08	270.00

* These heights are above the pole base plate.

** These are directions toward which the wind is flowing.
They are angles from the +X axis in the X-Y plane.

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Pole: Resistances (Extreme I)

Section Height* (ft)	Comb. Force Inter.	Applied Force				Factored Resistance			
		Axial (lb)	Shear (lb)	Bend. (ft-lb)	Torsion (ft-lb)	Axial $\phi=0.9$ (lb)	Shear $\phi=0.9$ (lb)	Bend. $\phi=0.9$ (ft-lb)	Torsion $\phi=0.95$ (ft-lb)
20.00	0.50	3,959.84	3,028.04	69,113.99	80,951.29	619,959.19	187,828.86	207,429.51	212,884.15
0.00	0.49	5,091.77	3,263.85	119,393.60	80,951.08	340,928.99	228,638.31	296,279.46	315,437.59

* These heights are above the pole base plate.

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Baseplate Analysis (Extreme I) - Pole1 - Pole

Combined Force Interaction	0.27
Critical Wind Direction *	220.00 deg
Alignment of Bend Line	135.00 deg
Width of Bending Section	16.527 (in)
Failure Line Start Coordinate (in)	(-11.500, 0.186)
Failure Line End Coordinate in	(0.186, -11.500)
Applied Bending Moment	19,097.24 ft-lb
Factored Bending Resistance	69,722.91 ft-lb
Plate Controlling Bolt Forces	

Bolt Number	Axial force (lb)	Moment Arm (in)	Bending moment (ft-lb)
3	-70513	3.25	-19097

Anchor Bolts Analysis (Extreme I) - Pole1 - Pole

Critical Wind Direct.* (deg)	Comb. Force Inter.	Applied Stress (psi)		Factored Resistance (psi)	
		Axial	Shear	$\phi F'_{nt}$	ϕF_v
220.00	0.42	17,734.28	375.28	42,187.50	22,500.00

* Per AISC Design Guide 1

* These are directions toward which the wind is flowing

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Pole Deflection Information: (Extreme I)

Critical Wind Direction: 270.00

Height (ft)	X-Defl. (in)	Y-Defl. (in)	Deflection Angle - X (deg)	Deflection Angle - Y (deg)
22.0000	-2.05	-3.02	-0.90	1.15
20.0000	-1.68	-2.54	-0.90	1.15
18.5000	-1.42	-2.18	-0.81	1.08
17.0000	-1.18	-1.85	-0.73	1.00
15.3333	-0.94	-1.52	-0.64	0.91
13.6667	-0.74	-1.21	-0.56	0.82
12.0000	-0.56	-0.94	-0.48	0.73
10.3333	-0.41	-0.70	-0.40	0.63
8.6667	-0.28	-0.49	-0.33	0.54
7.0000	-0.18	-0.32	-0.26	0.44
5.3333	-0.10	-0.19	-0.19	0.33
3.6667	-0.05	-0.09	-0.13	0.23
2.0000	-0.01	-0.03	-0.07	0.13
0.0000	0.00	0.00	0.00	0.00

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SERVICE I LIMIT STATE

Wind Velocity	76.0 mph
Dead Component Load Factor	1.00
Wind Load Factor	1.00
Gust Factor	1.14

Mast Arm 1: Wind and Weight Force Data (Service I)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft2)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)	Notes
ATTCHMT. 1	20.0000	43.0000	15.00	1.23	0.90	0.85	12.86	237	
ATTCHMT. 2	20.0000	55.0000	8.67	1.20	0.90	0.85	12.86	134	
ATTCHMT. 3	20.0000	63.0000	8.67	1.20	0.90	0.85	12.86	134	

Mast Arm 2: Wind and Weight Force Data (Service I)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft2)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)	Notes
ATTCHMT. 4	20.0000	20.0000	8.67	1.20	0.90	0.85	12.86	134	
ATTCHMT. 5	20.0000	26.0000	9.75	1.20	0.90	0.85	12.86	150	
ATTCHMT. 6	20.0000	32.0000	8.67	1.20	0.90	0.85	12.86	134	
ATTCHMT. 7	20.0000	38.0000	7.50	1.19	0.90	0.85	12.86	115	
ATTCHMT. 8	20.0000	44.0000	11.20	1.20	0.90	0.85	12.86	173	

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Arm: Forces and Moments (Service I)

Arm Type	Arm No.	Analysis Location	Forces (lb)			Moment (ft-lb)		
			Axial	Fy	Fz	Torsion	My	Mz
SIGNAL	1	BASE	44.53	930.91	1,910.01	1.86	49,494.32	37,826.60
SIGNAL	1	SPLICE-I	45.66	764.82	1,135.02	0.25	24,022.84	23,386.99
SIGNAL	1	SPLICE-O	42.64	762.20	956.67	0.01	21,204.80	21,329.70
SIGNAL	2	BASE	34.62	1,070.44	1,622.77	0.90	38,087.58	31,682.69
SIGNAL	2	SPLICE-I	36.62	755.07	858.99	0.02	12,545.76	11,596.69
SIGNAL	2	SPLICE-O	34.79	752.60	720.96	0.06	10,515.90	9,659.70

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Arm: Resistances (Service I)

Analysis Location			Comb. Force Inter.	Applied Forces			Factored Resistance Forces		
Arm Type	Arm No.	Site		Axial (lb)	Shear (lb)	Bend. (ft-lb)	Axial $\phi=0.9$ (lb)	Shear $\phi=0.9$ (lb)	Bend. $\phi=0.9$ (ft-lb)
SIGNAL	1	BASE	0.24	45	2,125	62,294	713,545	214,064	262,762
SIGNAL	1	SPLICE-I	0.18	46	1,369	33,527	598,958	179,687	191,565
SIGNAL	1	SPLICE-O	0.30	43	1,223	30,077	346,855	104,057	101,816
SIGNAL	2	BASE	0.26	35	1,944	49,542	553,834	166,150	191,534
SIGNAL	2	SPLICE-I	0.13	37	1,144	17,084	442,731	132,819	127,171
SIGNAL	2	SPLICE-O	0.16	35	1,042	14,279	319,468	95,840	87,354

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Arm Connection Analysis (Service I)

Analysis of Signal/Sign Arm Simplex Bolts

Mast Arm	Max Bolt CFI	Applied Forces (kip)		Factored Resistance (kip)	
		Tension	Shear	Tension	Shear
1	0.10	27.59	0.53	89.54	44.77
2	0.06	22.04	0.49	89.54	44.77

Analysis of Signal/Sign Arm Simplex Plates

Member Type	Max CSR	Applied Moment (ft-lb)	Factored Resistance (ft-lb)	Angle of Failure Line (deg)	Length of Bend Line (in)
1	0.12	13,580.60	114,733.70	45	18.89
2	0.10	11,311.42	117,779.92	45	19.39

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Pole: Wind and Weight Force Data (Service I)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft2)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)
ATTCHMT. 1	20.0000	43.0000	9.64	1.23	0.90	0.85	12.86	152
ATTCHMT. 2	20.0000	55.0000	8.67	1.20	0.90	0.85	12.86	134
ATTCHMT. 3	20.0000	63.0000	8.67	1.20	0.90	0.85	12.86	134
ATTCHMT. 4	20.0000	20.0000	8.67	1.20	0.90	0.85	12.86	134
ATTCHMT. 5	20.0000	26.0000	7.47	1.20	0.90	0.85	12.86	115
ATTCHMT. 6	20.0000	32.0000	8.67	1.20	0.90	0.85	12.86	134
ATTCHMT. 7	20.0000	38.0000	5.75	1.19	0.90	0.85	12.86	88
ATTCHMT. 8	20.0000	44.0000	11.20	1.20	0.90	0.85	12.86	173
22.0000	20.9964	0.0000	2.18	0.45	0.91	0.85	12.99	13
20.0000	19.2480	0.0000	1.66	0.45	0.89	0.85	12.76	10
18.5000	17.7481	0.0000	1.69	0.45	0.88	0.85	12.54	10
17.0000	16.1643	0.0000	1.91	0.45	0.86	0.85	12.29	11
15.3333	14.4977	0.0000	1.94	0.45	0.86	0.85	12.27	11
13.6667	12.8311	0.0000	1.97	0.45	0.86	0.85	12.27	11
12.0000	11.1644	0.0000	2.01	0.45	0.86	0.85	12.27	11
10.3333	9.4978	0.0000	2.04	0.45	0.86	0.85	12.27	11
8.6667	7.8312	0.0000	2.07	0.45	0.86	0.85	12.27	11
7.0000	6.1645	0.0000	2.10	0.45	0.86	0.85	12.27	12
5.3333	4.4979	0.0000	2.13	0.45	0.86	0.85	12.27	12
3.6667	2.8313	0.0000	2.17	0.45	0.86	0.85	12.27	12
2.0000	0.9971	0.0000	2.64	0.45	0.86	0.85	12.27	15

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
IN ACCORDANCE WITH AASHTO-2015 RQMTS. (FINAL DEFLECTED POSITION)

BY: SE70

08/18/2026

VERSION: 29.0.17.35

SUBJECT: JIMMY CARTER BLVD-POLE B-P22' M65'&55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P22M6555A15115

Pole: Forces and Moments (Service I)

Section Height* (ft)	Shear (lb)		Axial (lb)	Moment (ft-lb)		Torsion (ft-lb)	Wind Direct** (deg)
	Fx	Fy	Fz	Mx	My	Mz	
20.00	-1,054.97	-883.34	3,625.55	38,890.23	-50,276.76	-2,290.15	220.00
0.00	-1,103.29	-925.66	4,628.97	56,961.19	-71,855.35	-2,290.16	220.00

* These heights are above the pole base plate.

** These are directions toward which the wind is flowing.

They are angles from the +X axis in the X-Y plane.

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
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SUBJECT: JIMMY CARTER BLVD-POLE B-P22' M65'&55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P22M6555A15115

Pole: Resistances (Service I)

Section Height* (ft)	Comb. Force Inter.	Applied Force				Factored Resistance			
		Axial (lb)	Shear (lb)	Bend. (ft-lb)	Torsion (ft-lb)	Axial $\phi=0.9$ (lb)	Shear $\phi=0.9$ (lb)	Bend. $\phi=0.9$ (ft-lb)	Torsion $\phi=0.95$ (ft-lb)
20.00	0.31	3,625.55	1,375.95	63,562.58	2,290.15	619,959.19	187,828.86	207,429.51	212,884.15
0.00	0.32	4,628.97	1,440.17	91,693.89	2,290.16	340,928.99	228,638.31	296,279.46	315,437.59

* These heights are above the pole base plate.

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Folder: GAGWIN

File: P22M6555A15115

Baseplate Analysis (Service I) - Pole1 - Pole

Combined Force Interaction	0.19
Critical Wind Direction *	220.00 deg
Alignment of Bend Line	135.00 deg
Width of Bending Section	16.527 (in)
Failure Line Start Coordinate (in)	(-11.500, 0.186)
Failure Line End Coordinate in	(0.186, -11.500)
Applied Bending Moment	13,493.56 ft-lb
Factored Bending Resistance	69,722.91 ft-lb
Plate Controlling Bolt Forces	

Bolt Number	Axial force (lb)	Moment Arm (in)	Bending moment (ft-lb)
3	-49822	3.25	-13494

Anchor Bolts Analysis (Service I) - Pole1 - Pole

Critical Wind Direct.* (deg)	Comb. Force Inter.	Applied Stress (psi)		Factored Resistance (psi)	
		Axial	Shear	φ F'nt	φ Fv
220.00	0.30	12,530.53	180.52	42,187.50	22,500.00

* Per AISC Design Guide 1

* These are directions toward which the wind is flowing

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Pole Deflection Information: (Service I)

Critical Wind Direction: 220.00

Height (ft)	X-Defl. (in)	Y-Defl. (in)	Deflection Angle - X (deg)	Deflection Angle - Y (deg)
22.0000	-2.37	-1.86	-0.97	0.76
20.0000	-1.96	-1.54	-0.97	0.76
18.5000	-1.67	-1.31	-0.89	0.70
17.0000	-1.40	-1.10	-0.81	0.64
15.3333	-1.13	-0.89	-0.72	0.57
13.6667	-0.89	-0.70	-0.64	0.50
12.0000	-0.68	-0.54	-0.56	0.44
10.3333	-0.50	-0.40	-0.47	0.37
8.6667	-0.35	-0.28	-0.39	0.31
7.0000	-0.23	-0.18	-0.31	0.25
5.3333	-0.13	-0.10	-0.24	0.19
3.6667	-0.06	-0.05	-0.16	0.13
2.0000	-0.02	-0.01	-0.09	0.07
0.0000	0.00	0.00	0.00	0.00

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BY: SE70

08/18/2026

VERSION: 29.0.17.35

SUBJECT: JIMMY CARTER BLVD-POLE B-P22' M65'x55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P22M6555A15115

ICE LIMIT STATE

Wind Velocity	76.0 mph
Dead Component Load Factor	1.10
Wind Load Factor	1.00
Gust Factor	1.14

Mast Arm 1: Wind and Weight Force Data (Ice)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft ²)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)	Notes
ATTCHMT. 1	20.0000	43.0000	15.00	1.23	0.90	0.85	12.86	237	
ATTCHMT. 2	20.0000	55.0000	8.67	1.20	0.90	0.85	12.86	134	
ATTCHMT. 3	20.0000	63.0000	8.67	1.20	0.90	0.85	12.86	134	

Mast Arm 2: Wind and Weight Force Data (Ice)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft ²)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)	Notes
ATTCHMT. 4	20.0000	20.0000	8.67	1.20	0.90	0.85	12.86	134	
ATTCHMT. 5	20.0000	26.0000	9.75	1.20	0.90	0.85	12.86	150	
ATTCHMT. 6	20.0000	32.0000	8.67	1.20	0.90	0.85	12.86	134	
ATTCHMT. 7	20.0000	38.0000	7.50	1.19	0.90	0.85	12.86	115	
ATTCHMT. 8	20.0000	44.0000	11.20	1.20	0.90	0.85	12.86	173	

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File: P22M6555A15115

Arm: Forces and Moments (Ice)

Arm Type	Arm No.	Analysis Location	Forces (lb)			Moment (ft-lb)		
			Axial	Fy	Fz	Torsion	My	Mz
SIGNAL	1	BASE	104.54	975.26	3,052.12	2.17	90,466.75	39,190.61
SIGNAL	1	SPLICE-I	111.52	794.44	1,999.38	0.30	48,156.26	24,130.62
SIGNAL	1	SPLICE-O	103.89	788.46	1,757.17	0.01	43,095.09	21,997.96
SIGNAL	2	BASE	84.07	1,124.64	2,842.59	1.03	72,097.79	33,087.88
SIGNAL	2	SPLICE-I	90.64	786.63	1,629.61	0.06	24,913.32	12,084.32
SIGNAL	2	SPLICE-O	85.40	780.77	1,433.23	0.11	20,978.19	10,070.52

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VERSION: 29.0.17.35

SUBJECT: JIMMY CARTER BLVD-POLE B-P22' M65'&55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P22M6555A15115

Arm: Resistances (Ice)

Analysis Location			Comb. Force Inter.	Applied Forces			Factored Resistance Forces		
Arm Type	Arm No.	Site		Axial (lb)	Shear (lb)	Bend. (ft-lb)	Axial $\phi=0.9$ (lb)	Shear $\phi=0.9$ (lb)	Bend. $\phi=0.9$ (ft-lb)
SIGNAL	1	BASE	0.38	105	3,204	98,591	713,545	214,064	262,762
SIGNAL	1	SPLICE-I	0.28	112	2,151	53,864	598,958	179,687	191,565
SIGNAL	1	SPLICE-O	0.48	104	1,926	48,385	346,855	104,057	101,816
SIGNAL	2	BASE	0.41	84	3,057	79,328	553,834	166,150	191,534
SIGNAL	2	SPLICE-I	0.22	91	1,810	27,689	442,731	132,819	127,171
SIGNAL	2	SPLICE-O	0.27	85	1,632	23,270	319,468	95,840	87,354

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Folder: GAGWIN

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Arm Connection Analysis (Ice)

Analysis of Signal/Sign Arm Simplex Bolts

Mast Arm	Max Bolt CFI	Applied Forces (kip)		Factored Resistance (kip)	
		Tension	Shear	Tension	Shear
1	0.21	40.97	0.80	89.54	44.77
2	0.14	33.24	0.76	89.54	44.77

Analysis of Signal/Sign Arm Simplex Plates

Member Type	Max CSR	Applied Moment (ft-lb)	Factored Resistance (ft-lb)	Angle of Failure Line (deg)	Length of Bend Line (in)
1	0.18	20,169.70	114,733.70	45	18.89
2	0.14	17,057.19	117,779.92	45	19.39

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SUBJECT: JIMMY CARTER BLVD-POLE B-P22' M65'&55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P22M6555A15115

Pole: Wind and Weight Force Data (Ice)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft2)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)
ATTCHMT. 1	20.0000	43.0000	15.00	1.23	0.90	0.85	12.86	237
ATTCHMT. 2	20.0000	55.0000	8.67	1.20	0.90	0.85	12.86	134
ATTCHMT. 3	20.0000	63.0000	8.67	1.20	0.90	0.85	12.86	134
ATTCHMT. 4	20.0000	20.0000	8.67	1.20	0.90	0.85	12.86	134
ATTCHMT. 5	20.0000	26.0000	0.00	1.20	0.90	0.85	12.86	0
ATTCHMT. 6	20.0000	32.0000	8.67	1.20	0.90	0.85	12.86	134
ATTCHMT. 7	20.0000	38.0000	0.00	1.19	0.90	0.85	12.86	0
ATTCHMT. 8	20.0000	44.0000	11.20	1.20	0.90	0.85	12.86	173
22.0000	20.9964	0.0000	2.18	0.45	0.91	0.85	12.99	13
20.0000	19.2480	0.0000	1.66	0.45	0.89	0.85	12.76	10
18.5000	17.7481	0.0000	1.69	0.45	0.88	0.85	12.54	10
17.0000	16.1643	0.0000	1.91	0.45	0.86	0.85	12.29	11
15.3333	14.4977	0.0000	1.94	0.45	0.86	0.85	12.27	11
13.6667	12.8311	0.0000	1.97	0.45	0.86	0.85	12.27	11
12.0000	11.1644	0.0000	2.01	0.45	0.86	0.85	12.27	11
10.3333	9.4978	0.0000	2.04	0.45	0.86	0.85	12.27	11
8.6667	7.8312	0.0000	2.07	0.45	0.86	0.85	12.27	11
7.0000	6.1645	0.0000	2.10	0.45	0.86	0.85	12.27	12
5.3333	4.4979	0.0000	2.13	0.45	0.86	0.85	12.27	12
3.6667	2.8313	0.0000	2.17	0.45	0.86	0.85	12.27	12
2.0000	0.9971	0.0000	2.64	0.45	0.86	0.85	12.27	15

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
IN ACCORDANCE WITH AASHTO-2015 RQMTS. (FINAL DEFLECTED POSITION)

BY: SE70

08/18/2026

VERSION: 29.0.17.35

SUBJECT: JIMMY CARTER BLVD-POLE B-P22' M65'&55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P22M6555A15115

Pole: Forces and Moments (Ice)

Section Height* (ft)	Shear (lb)		Axial (lb)	Moment (ft-lb)		Torsion (ft-lb)	Wind Direct** (deg)
	Fx	Fy	Fz	Mx	My	Mz	
20.00	-170.07	-1,495.89	6,035.53	73,991.97	-92,139.40	37,559.77	270.00
0.00	-8.19	-1,496.14	7,397.18	104,134.27	-93,633.51	37,559.63	270.00

* These heights are above the pole base plate.

** These are directions toward which the wind is flowing.
They are angles from the +X axis in the X-Y plane.

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Folder: GAGWIN

File: P22M6555A15115

Pole: Resistances (Ice)

Section Height* (ft)	Comb. Force Inter.	Applied Force				Factored Resistance			
		Axial (lb)	Shear (lb)	Bend. (ft-lb)	Torsion (ft-lb)	Axial $\phi=0.9$ (lb)	Shear $\phi=0.9$ (lb)	Bend. $\phi=0.9$ (ft-lb)	Torsion $\phi=0.95$ (ft-lb)
20.00	0.57	6,035.53	1,505.52	118,171.40	37,559.77	619,959.19	187,828.86	207,429.51	212,884.15
0.00	0.48	7,397.18	1,496.16	140,039.92	37,559.63	340,928.99	228,638.31	296,279.46	315,437.59

* These heights are above the pole base plate.

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Folder: GAGWIN

File: P22M6555A15115

Baseplate Analysis (Ice) - Pole1 - Pole

Combined Force Interaction	0.31
Critical Wind Direction *	220.00 deg
Alignment of Bend Line	135.00 deg
Width of Bending Section	16.527 (in)
Failure Line Start Coordinate (in)	(-11.500, 0.186)
Failure Line End Coordinate in	(0.186, -11.500)
Applied Bending Moment	21,629.30 ft-lb
Factored Bending Resistance	69,722.91 ft-lb
Plate Controlling Bolt Forces	

Bolt Number	Axial force (lb)	Moment Arm (in)	Bending moment (ft-lb)
3	-79862	3.25	-21629

Anchor Bolts Analysis (Ice) - Pole1 - Pole

Critical Wind Direct.* (deg)	Comb. Force Inter.	Applied Stress (psi)		Factored Resistance (psi)	
		Axial	Shear	ϕ F'nt	ϕ Fv
220.00	0.48	20,085.63	179.95	42,187.50	22,500.00

* Per AISC Design Guide 1

* These are directions toward which the wind is flowing

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Pole Deflection Information: (Ice)

Critical Wind Direction: 270.00

Height (ft)	X-Defl. (in)	Y-Defl. (in)	Deflection Angle - X (deg)	Deflection Angle - Y (deg)
22.0000	-3.48	-3.47	-1.51	1.42
20.0000	-2.85	-2.87	-1.51	1.42
18.5000	-2.40	-2.44	-1.36	1.30
17.0000	-2.00	-2.04	-1.22	1.19
15.3333	-1.60	-1.65	-1.08	1.06
13.6667	-1.25	-1.30	-0.93	0.93
12.0000	-0.95	-1.00	-0.80	0.81
10.3333	-0.69	-0.73	-0.67	0.69
8.6667	-0.48	-0.51	-0.55	0.57
7.0000	-0.31	-0.33	-0.43	0.46
5.3333	-0.18	-0.19	-0.32	0.35
3.6667	-0.08	-0.09	-0.22	0.23
2.0000	-0.02	-0.03	-0.12	0.13
0.0000	0.00	0.00	0.00	0.00

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Folder: GAGWIN

File: P22M6555A15115

FATIGUE II LIMIT STATE

Galloping	No
Natural Wind Gust (11.2 mph)	Yes
Truck-Induced Gust (65.0 mph)	No
Importance Factor	II

Mast Arm: Fatigue Analysis (Fatigue II)

Analysis Location			Design Load	Comb. Force Inter.	Moment (ft-lb)	Shear force (lb)	Shear Stress (ksi)	Applied Bending Stress (ksi)	Allowable Stress (ksi)
Arm Type	Arm No.	Site							
MA	1	BASE	NATURAL WIND GUST	0.80	15,795.45	427	0.06	3.58	4.50
MA	1	SP-I	NATURAL WIND GUST	0.25	9,357.68	325	0.05	3.01	12.00
MA	1	SP-O	NATURAL WIND GUST	0.39	8,481.49	325	0.09	4.67	12.00
MA	2	BASE	NATURAL WIND GUST	0.87	12,926.83	455	0.08	3.90	4.50
MA	2	SP-I	NATURAL WIND GUST	0.19	4,709.09	299	0.07	2.22	12.00
MA	2	SP-O	NATURAL WIND GUST	0.21	3,940.56	299	0.09	2.56	12.00

Mast Arm: Deflections (Fatigue II)

Arm Type	Arm No.	Load Case	Max Vertical Deflection (in)
MA	1	NATURAL WIND GUST	1.17
MA	2	NATURAL WIND GUST	1.04

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Fatigue Analysis of Signal and Sign / Pole Connection: Arm

Arm Type	Arm No.	Component	Load	Stress Ratio	Applied Stress (ksi)	Allowable Stress (ksi)
MA	1	SIMPLEX BOLT	NATURAL WIND GUST	0.74	5.16	7.00
MA	2	SIMPLEX BOLT	NATURAL WIND GUST	0.60	4.22	7.00

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Analysis of Pole (Fatigue II)

Section Height* (ft)	Design Load	Comb. Force Inter.	Moment (ft-lb)	Applied Bending Stress (ksi)	Allowable Bending Stress (ksi)	Deflection (in)
0.00	NATURAL WIND GUST	0.56	12,664.76	2.52	4.50	0.00

* These heights are above the pole base plate.

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Tube to Transverse Plate CAFT Calculation Details

	Pole to Baseplate Weld	Arm M65 Shaft to Simplex Plate Weld	Arm 55M Shaft to Simplex Plate Weld
Weld type	Socket	Socket	Socket
Tt (in)	0.31250	0.31250	0.25000
Dt (in)	16.00	15.00	14.50
Ttp (in)	2.50	3.00	3.00
Dbc (in)	22.50	26.87	26.87
CBC	1.41	1.79	1.85
Dop (in)	N/A	N/A	N/A
COP	N/A	N/A	N/A
NS	0.00	0.00	0.00
RRb (in)	N/A	N/A	N/A
Multisided Factor	N/A	N/A	N/A
Kf	2.77	2.75	2.67
Ki	6.26	6.21	5.69
CAFT (ksi)	4.50	4.50	4.50

NOTE: The maximum bolt circle is used for bolt patterns where all the bolts do not lie on a single circle, per AASHTO.

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
IN ACCORDANCE WITH AASHTO-2015 RQMTS. (FINAL DEFLECTED POSITION)

BY: SE70

08/18/2026

VERSION: 29.0.17.35

SUBJECT: JIMMY CARTER BLVD-POLE B-P22' M65'&55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P22M6555A15115

Fatigue Analysis of Anchor Bolts (NATURAL WIND GUST)

Load Case	Combined Stress Ratio	Axial (lb)
NATURAL WIND GUST	0.26	5,968.47

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SUBJECT: JIMMY CARTER BLVD-POLE B-P22' M65'x55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P22M6555A15115

Opening Group on the Pole

Description	Attachment Height (ft)	Clear Opening (in)			Reinforcement (in)		
		Width	Height	Inside Corner Radius	Rim Thickness	Rim Depth	Rim Projection
STD HH	2.75	4.48	7.00	2.24	0.28	2.50	0.50

Description	Location On Pole (ft)	Orientation (deg)	Tube Diam. (in)	Tube Thick. (in)	Area (in ²)	X Centroid (in)	Y Centroid (in)	Ix(in ⁴)	Iy(in ⁴)
STD HH	2.75	0.00	15.62	0.31	14.89	-0.14	0.00	448	424

		Stress at Root		Stress at Toe		
Description	Moment (ft-lb)	Actual (ksi)	Resist (ksi)	Actual (ksi)	Resist (ksi)	Max CSR
NATURAL WIND GUST	10,319.67	8.86	16.00	2.21	7.00	0.55

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BY: SE70

08/18/2026

VERSION: 29.0.17.35

SUBJECT: JIMMY CARTER BLVD-POLE C-P30' M55'&M55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P30M5555A15115

Design Criteria

Design Code	AASHTO-2015	Fatigue Category	2
Ultimate Wind Speed (mph)	115.0	Truck Gust	No
Mean Recurrence Interval	700	Galloping	No
Service Level Wind Speed (mph)	76.0	Natural Wind Gust	Yes
AASHTO Ice Included ?	Yes		

Design Summary - Pole

Height (ft)	Shaft Weight (lb)	Ground Line Diameter (in)	Top Dia. (in)
30.0000	1361	16.00	11.800

Section Characteristics

	Section - 1
Shape	Round
Top Dia. (in)	11.800
Base Diameter (in)	16.000
Thickness (in)	0.31250
Length (ft)	30.00
Shaft Weight (lb)	1361
Assembly Weight (lb)	1898
Taper (in/ft)	0.14000
Yield Strength (ksi)	55.00
Material	S220 - A572

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SUBJECT: JIMMY CARTER BLVD-POLE C-P30' M55'&M55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P30M5555A15115

Base Plate

Shape	Square
Material	S70 - A36
Width (in)	23.000
Thickness (in)	2.50000
Yield Strength (ksi)	36.00
Base Weld Type	SOCKET
Weight (lb)	207

Anchor Bolts

Material	S100 - F1554
Bolt diameter (in)	2.25
Bolt circle diameter (in)	22.50
Quantity	4
Yield Strength (ksi)	55.00
Tensile strength (ksi)	75

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SUBJECT: JIMMY CARTER BLVD-POLE C-P30' M55'&M55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P30M5555A15115

Design Summary - Arms

Signal and Sign	Arm 1	Arm 2
Shape	Round	Round
Span Length (ft)	55.0000	55.0000
Taper (in/ft)	0.14000	0.14000
Attachment Height (ft)	20.00	20.00
Orientation (deg)	180.00	277.00
Slope at Base (deg)	0.00	0.00
Centroid Location		
Horizontal (ft)	24.3929	24.3929
Above Attachment (ft)	0.0000	0.0000
Unbent Length (ft)	55.0000	55.0000
Material-Base	S220 - 55 ksi	S220 - 55 ksi
Weight (lb)	1399	1399
Base Section		
Base O.D. (in)	14.50	14.50
Thickness (in)	0.25000	0.25000
Length (ft)	22.9880	22.9880
Yield Strength (ksi)	55.00	55.00
Material	S220	S220
Joint Type	Slip Joint	Slip Joint
Overlap Length (ft)	2.5695	2.5695
Outer Section		
Base O.D. (in)	12.00	12.00
Thickness (in)	0.17930	0.17930
Length (ft)	34.5814	34.5814
Yield Strength (ksi)	55.00	55.00
Material	S105	S105

Base Weld Type = Socket

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BY: SE70
SUBJECT: JIMMY CARTER BLVD-POLE C-P30' M55'&M55' AASHTO 2015 115MPH
Folder: GAGWIN
Design Summary - Arms

VERSION: 29.0.17.35

File: P30M5555A15115

Simplex Dimensions

	Arm 1	Arm 2
Connection Bolt Data		
Number of bolts	4	4
Bolt diameter (in)	1.25	1.25
ASTM Specification	A325	A325
Horizontal Spacing (in)	19.00	19.00
Vertical Spacing (in)	19.00	19.00
Attachment Plate Data		
Horizontal Width (in)	24.00	24.00
Vertical Width (in)	24.00	24.00
Mast Arm Bracket Thickness (in)	3.00	3.00
Arm Plate Bracket Weight (lb)	247	247
Pole Plate Bracket Thickness (in)	2.00	2.00
Pole Plate Bracket Weight (lb)	165	165
Yield Strength (ksi)	36.00	36.00
Vertical Gusset Thickness (in)	0.5000	0.5000
Horizontal Gusset Thickness (in)	0.5000	0.5000

Attachment Type

Arm 1:	SIMPLEX - THRU BOLTS,	Base Weld Type = Socket
Arm 2:	SIMPLEX - THRU BOLTS,	Base Weld Type = Socket

** These heights are above bottom of base plate or transformer base.

Elliptical cross section; first diameter is horizontal.

** Arm orientations are angles from +X axis in X-Y plane.

X and Y axes are perpendicular/parallel to sides of pole base plate. See *** below.

*** If arm is attached with a clamp, height and orientation must not be changed from values shown above without consulting Valmont.

Nice to have:

**** Assembly weight includes unfinished shaft + flange + simplex plate.

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Description of EPA Loading

Description of Load	Position of Load	Mounting Height ** (ft)	Centroid Height ** (ft)	Distance To Centroid From Pole (ft)	Weight (lb)	Effective Projected Area (ft2)
CCTV CAMERA	Pole	30.0000	30.0000	1.0000	10	2.00

Description of Sign Loading

Position of Signal or Sign	Mounting Height ** (ft)	Centroid Height ** (ft)	Distance To Centroid From Pole (ft)	Sign Weight (lb)	Sign Width (ft)	Sign Height (ft)	Sign Depth (ft)	Sign Cd
Mast Arm 1	20.0000	20.0000	45.0000	30	10.0000	1.500	0.000	1.23
Mast Arm 2	20.0000	20.0000	30.0000	20	6.5000	1.500	0.000	1.20
Mast Arm 2	20.0000	20.0000	42.0000	15	2.5000	3.000	0.000	1.19

Description of Luminaire/Signal Loading

Position of Signal	Mounting Height ** (ft)	Centroid Height ** (ft)	Distance To Centroid From Pole (ft)	Weight (lb)	Vertical Plane Area (ft2)	Horizontal Plane Area (ft2)
Arm 1	20.0000	20.0000	39.0000	50	8.67	1.80
Arm 1	20.0000	20.0000	51.0000	50	8.67	1.80
Arm 2	20.0000	20.0000	24.0000	50	8.67	1.80
Arm 2	20.0000	20.0000	36.0000	50	8.67	1.80
Arm 2	20.0000	20.0000	48.0000	62	11.20	1.80

THE VALUES SHOWN IN THIS TABLE MUST NOT BE EXCEEDED WITHOUT CONSULTING VALMONT.

ANY SIZES OR OTHER DIMENSIONS NOT PROVIDED BY THE SPECIFYING AGENCY HAVE BEEN ESTIMATED BY VALMONT.

** THESE HEIGHTS ARE ABOVE BOTTOM OF BASE PLATE OR TRANSFORMER BASE.

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
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08/18/2026

VERSION: 29.0.17.35

SUBJECT: JIMMY CARTER BLVD-POLE C-P30' M55'&M55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P30M5555A15115

RESULTS SUMMARY - Pole

Maximum Combined Force Interaction In Each Major Component		Maximum Reactions Applied To Foundation	
Strength I		Bending Moment	129568 ft-lb
Pole CFI (At 20.00 (ft))	0.32	Torsion	71643 ft-lb
Signal and Sign Arm 1 CFI	0.25	Shear Force	3459 lb
Signal and Sign Arm 2 CFI	0.26	Axial Force	7419 lb
Base Plate CFI	0.14		
Anchor Bolts CFI	0.22	Ice	
S/S Arm 1 Att. Bolts CFI	0.03	Pole CFI (At 20.00 (ft))	0.48
S/S Arm 1 Att. Plate CFI	0.07	Signal and Sign Arm 1 CFI	0.40
S/S Pole 1 Att. Plate CFI	0.15	Signal and Sign Arm 2 CFI	0.43
S/S Arm 2 Att. Bolts CFI	0.03	Base Plate CFI	0.27
S/S Arm 2 Att. Plate CFI	0.07	Anchor Bolts CFI	0.42
S/S Pole 2 Att. Plate CFI	0.15	S/S Arm 1 Att. Bolts CFI	0.13
Deflection % (At 30.00 (ft))	1.191 %	S/S Arm 1 Att. Plate CFI	0.14
Deflection (At 30.00 (ft))	4.29 in	S/S Pole 1 Att. Plate CFI	0.32
Rotation (At 30.00 (ft))	1.07 deg	S/S Arm 2 Att. Bolts CFI	0.15
		S/S Arm 2 Att. Plate CFI	0.15
Extreme I		S/S Pole 2 Att. Plate CFI	0.34
Pole CFI (At 0.00 (ft))	0.47	Deflection % (At 30.00	1.976 %
Signal and Sign Arm 1 CFI	0.42	Deflection (At 30.00 (ft))	7.11 in
Signal and Sign Arm 2 CFI	0.44	Rotation (At 30.00 (ft))	1.76 deg
Base Plate CFI	0.26		
Anchor Bolts CFI	0.40	Fatigue II	
S/S Arm 1 Att. Bolts CFI	0.15	Pole (At 0.00 (ft))	0.58
S/S Arm 1 Att. Plate CFI	0.15	Hand Hole / Opening	0.63
S/S Pole 1 Att. Plate CFI	0.34	Signal and Sign Arm 1	
S/S Arm 2 Att. Bolts CFI	0.17	Arm Tube	0.86
S/S Arm 2 Att. Plate CFI	0.16	Simplex Bolts	0.60
S/S Pole 2 Att. Plate CFI	0.36	Signal and Sign Arm 2	
Deflection % (At 30.00 (ft))	1.584 %	Arm Tube	0.89
Deflection (At 30.00 (ft))	5.70 in	Simplex Bolts	0.62
Rotation (At 30.00 (ft))	1.35 deg	Anchor Bolts	0.30
Service I			
Pole CFI (At 0.00 (ft))	0.29		
Signal and Sign Arm 1 CFI	0.26		
Signal and Sign Arm 2 CFI	0.27		
Base Plate CFI	0.18		
Anchor Bolts CFI	0.27		
S/S Arm 1 Att. Bolts CFI	0.06		
S/S Arm 1 Att. Plate CFI	0.10		
S/S Pole 1 Att. Plate CFI	0.21		
S/S Arm 2 Att. Bolts CFI	0.07		
S/S Arm 2 Att. Plate CFI	0.10		
S/S Pole 2 Att. Plate CFI	0.22		
Deflection % (At 30.00 (ft))	1.231 %		
Deflection (At 30.00 (ft))	4.43 in		
Rotation (At 30.00 (ft))	1.07 deg		

SHAFT PROPERTIES: NUMER OF SIDES = Round
DIAMETER (D) = 16.000 IN
THICKNESS (T) = 0.31250 IN
YIELD STRENGTH (KSI) = 55
MOM. OF INERTIA (IX) = 473.138 IN^4
SECTION MODULUS (SX) = 59.142 IN^3

MN: ALLOWABLE SHAFT MOMENT

$$MB = (Fb * Sx) / 12$$

$$= (55000 * 59.142 / 12 = 271069$$

MB = 271069 FT-LB

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
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BY: SE70

08/18/2026

VERSION: 29.0.17.35

SUBJECT: JIMMY CARTER BLVD-POLE C-P30' M55'&M55' AASHTO 2015 115MPH

Folder: GAGWIN
Pole Properties

File: P30M5555A15115

Height (ft)	Diameter (in)	Wall Thk. (in)	Roundness Ratio (%)	D/t	B/T	Moments of Inertia (in4)	Plastic Section Modulus (in3)	Area (in2)	Radius of Gyration (in)
30.0000	11.80	0.31250	100.0	37.76	0.00	185.80	41.20	11.27	4.06
27.5000	12.15	0.31250	100.0	38.88	0.00	203.30	43.75	11.62	4.18
25.0000	12.50	0.31250	100.0	40.00	0.00	221.87	46.38	11.96	4.31
22.5000	12.85	0.31250	100.0	41.12	0.00	241.54	49.08	12.30	4.43
20.0000	13.20	0.31250	100.0	42.24	0.00	262.33	51.86	12.65	4.55
17.5000	13.55	0.31250	100.0	43.36	0.00	284.29	54.71	12.99	4.68
15.0000	13.90	0.31250	100.0	44.48	0.00	307.44	57.64	13.34	4.80
12.5000	14.25	0.31250	100.0	45.60	0.00	331.81	60.65	13.68	4.93
10.0000	14.60	0.31250	100.0	46.72	0.00	357.44	63.73	14.02	5.05
7.5000	14.95	0.31250	100.0	47.84	0.00	384.36	66.90	14.37	5.17
5.0000	15.30	0.31250	100.0	48.96	0.00	412.59	70.13	14.71	5.30
2.5000	15.65	0.31250	100.0	50.08	0.00	442.17	73.45	15.05	5.42
0.0000	16.00	0.31250	100.0	51.20	0.00	473.14	76.84	15.40	5.54

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08/18/2026

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SUBJECT: JIMMY CARTER BLVD-POLE C-P30' M55'&M55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P30M5555A15115

Arm: Forces and Moments (Strength I)

Arm Type	Arm No.	Analysis Location	Forces (lb)			Moment (ft-lb)		
			Axial	Fy	Fz	Torsion	My	Mz
SIGNAL	1	BASE	28.37	28.12	1,968.26	0.00	47,605.27	680.24
SIGNAL	1	SPLICE-I	29.69	15.28	1,069.27	0.00	16,917.39	241.73
SIGNAL	1	SPLICE-O	26.15	12.81	896.74	0.00	14,391.55	205.64
SIGNAL	2	BASE	30.89	28.30	2,051.99	0.00	49,295.53	679.96
SIGNAL	2	SPLICE-I	33.10	15.90	1,152.96	0.00	16,898.42	233.09
SIGNAL	2	SPLICE-O	29.17	13.52	980.44	0.00	14,157.55	195.28

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Arm: Resistances (Strength I)

Analysis Location			Comb. Force Inter.	Applied Forces			Factored Resistance Forces		
Arm Type	Arm No.	Site		Axial (lb)	Shear (lb)	Bend. (ft-lb)	Axial $\phi=0.9$ (lb)	Shear $\phi=0.9$ (lb)	Bend. $\phi=0.9$ (ft-lb)
SIGNAL	1	BASE	0.25	28	1,968	47,610	553,834	166,150	191,534
SIGNAL	1	SPLICE-I	0.13	30	1,069	16,919	442,731	132,819	127,171
SIGNAL	1	SPLICE-O	0.16	26	897	14,393	319,468	95,840	87,354
SIGNAL	2	BASE	0.26	31	2,052	49,300	553,834	166,150	191,534
SIGNAL	2	SPLICE-I	0.13	33	1,153	16,900	442,731	132,819	127,171
SIGNAL	2	SPLICE-O	0.16	29	981	14,159	319,468	95,840	87,354

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Arm Connection Analysis (Strength I)

Analysis of Signal/Sign Arm Simplex Bolts

Mast Arm	Max Bolt CFI	Applied Forces (kip)		Factored Resistance (kip)	
		Tension	Shear	Tension	Shear
1	0.03	15.26	0.49	89.54	44.77
2	0.03	15.79	0.51	89.54	44.77

Analysis of Signal/Sign Arm Simplex Plates

Member Type	Max CSR	Applied Moment (ft-lb)	Factored Resistance (ft-lb)	Angle of Failure Line (deg)	Length of Bend Line (in)
1	0.07	7,828.80	117,779.92	45	19.39
2	0.07	8,103.00	117,779.92	45	19.39

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Pole: Forces and Moments (Strength I)

Section Height* (ft)	Shear (lb)		Axial (lb)	Moment (ft-lb)		Torsion (ft-lb)	Wind Direct** (deg)
	Fx	Fy	Fz	Mx	My	Mz	
30.00	-0.15	-0.18	12.50	0.00	12.50	-0.18	0.00
20.00	-50.88	-59.87	4,553.14	50,086.59	-42,562.72	0.00	0.00
0.00	-3.37	-3.97	5,773.58	50,723.89	-43,104.29	0.00	0.00

* These heights are above the pole base plate.

** These are directions toward which the wind is flowing.
They are angles from the +X axis in the X-Y plane.

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Folder: GAGWIN

File: P30M5555A15115

Pole: Resistances (Strength I)

Section Height* (ft)	Comb. Force Inter.	Applied Force				Factored Resistance			
		Axial (lb)	Shear (lb)	Bend. (ft-lb)	Torsion (ft-lb)	Axial $\phi=0.9$ (lb)	Shear $\phi=0.9$ (lb)	Bend. $\phi=0.9$ (ft-lb)	Torsion $\phi=0.95$ (ft-lb)
30.00	0.00	12.50	0.23	12.50	0.18	556,353.51	167,424.14	168,836.70	169,144.17
20.00	0.32	4,553.14	78.57	65,728.62	0.00	489,431.27	187,828.86	207,429.51	212,884.15
0.00	0.24	5,773.58	5.21	66,564.96	0.00	187,018.38	228,638.31	296,279.46	315,437.59

* These heights are above the pole base plate.

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Baseplate Analysis (Strength I) - Pole1 - Pole

Combined Force Interaction	0.14
Critical Wind Direction *	0.00 deg
Alignment of Bend Line	135.00 deg
Width of Bending Section	16.527 (in)
Failure Line Start Coordinate (in)	(-11.500, 0.186)
Failure Line End Coordinate in	(0.186, -11.500)
Applied Bending Moment	9,974.31 ft-lb
Factored Bending Resistance	69,722.91 ft-lb
Plate Controlling Bolt Forces	

Bolt Number	Axial force (lb)	Moment Arm (in)	Bending moment (ft-lb)
3	-36828	3.25	-9974

Anchor Bolts Analysis (Strength I) - Pole1 - Pole

Critical Wind Direct.* (deg)	Comb. Force Inter.	Applied Stress (psi)		Factored Resistance (psi)	
		Axial	Shear	φ F'nt	φ Fv
0.00	0.22	9,262.45	0.00	42,187.50	22,500.00

* Per AISC Design Guide 1

* These are directions toward which the wind is flowing

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Folder: GAGWIN

File: P30M5555A15115

Pole Deflection Information: (Strength I)

Critical Wind Direction: 0.00

Height (ft)	X-Defl. (in)	Y-Defl. (in)	Deflection Angle - X (deg)	Deflection Angle - Y (deg)
30.0000	-2.78	-3.27	-0.70	0.82
27.5000	-2.41	-2.84	-0.70	0.82
25.0000	-2.05	-2.41	-0.70	0.82
22.5000	-1.68	-1.98	-0.70	0.82
20.0000	-1.32	-1.55	-0.70	0.82
17.5000	-0.98	-1.16	-0.58	0.69
15.0000	-0.71	-0.83	-0.48	0.57
12.5000	-0.48	-0.56	-0.39	0.45
10.0000	-0.30	-0.35	-0.30	0.35
7.5000	-0.16	-0.19	-0.22	0.25
5.0000	-0.07	-0.08	-0.14	0.16
2.5000	-0.02	-0.02	-0.07	0.08
0.0000	0.00	0.00	0.00	0.00

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SUBJECT: JIMMY CARTER BLVD-POLE C-P30' M55'&M55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P30M5555A15115

EXTREME I LIMIT STATE

Wind Velocity	115.0 mph
Dead Component Load Factor	1.10
Wind Load Factor	1.00
Gust Factor	1.14

Mast Arm 1: Wind and Weight Force Data (Extreme I)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft2)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)	Notes
ATTCHMT. 1	20.0000	39.0000	8.67	1.20	0.90	0.85	29.44	306	
ATTCHMT. 2	20.0000	45.0000	15.00	1.23	0.90	0.85	29.44	543	
ATTCHMT. 3	20.0000	51.0000	8.67	1.20	0.90	0.85	29.44	306	

Mast Arm 2: Wind and Weight Force Data (Extreme I)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft2)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)	Notes
ATTCHMT. 4	20.0000	24.0000	8.67	1.20	0.90	0.85	29.44	306	
ATTCHMT. 5	20.0000	30.0000	9.74	1.20	0.90	0.85	29.44	344	
ATTCHMT. 6	20.0000	36.0000	8.67	1.20	0.90	0.85	29.44	306	
ATTCHMT. 7	20.0000	42.0000	7.50	1.19	0.90	0.85	29.44	263	
ATTCHMT. 8	20.0000	48.0000	11.20	1.20	0.90	0.85	29.44	396	

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
IN ACCORDANCE WITH AASHTO-2015 RQMTS. (FINAL DEFLECTED POSITION)

BY: SE70

08/18/2026

VERSION: 29.0.17.35

SUBJECT: JIMMY CARTER BLVD-POLE C-P30' M55'&M55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P30M5555A15115

Arm: Forces and Moments (Extreme I)

Arm Type	Arm No.	Analysis Location	Forces (lb)			Moment (ft-lb)		
			Axial	Fy	Fz	Torsion	My	Mz
SIGNAL	1	BASE	68.60	1,844.65	1,693.80	4.07	40,432.79	68,684.29
SIGNAL	1	SPLICE-I	94.86	1,500.19	909.10	0.44	14,147.75	34,274.67
SIGNAL	1	SPLICE-O	96.39	1,496.83	757.20	0.21	12,006.87	30,423.96
SIGNAL	2	BASE	9.86	2,232.85	1,761.24	3.37	41,902.80	73,724.91
SIGNAL	2	SPLICE-I	56.74	1,925.75	976.00	0.03	14,245.67	31,037.79
SIGNAL	2	SPLICE-O	57.38	1,922.63	824.16	0.02	11,932.88	26,093.53

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SUBJECT: JIMMY CARTER BLVD-POLE C-P30' M55'&M55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P30M5555A15115

Arm: Resistances (Extreme I)

Analysis Location			Comb. Force Inter.	Applied Forces			Factored Resistance Forces		
Arm Type	Arm No.	Site		Axial (lb)	Shear (lb)	Bend. (ft-lb)	Axial $\phi=0.9$ (lb)	Shear $\phi=0.9$ (lb)	Bend. $\phi=0.9$ (ft-lb)
SIGNAL	1	BASE	0.42	69	2,504	79,702	553,834	166,150	191,534
SIGNAL	1	SPLICE-I	0.29	95	1,754	37,080	442,731	132,819	127,171
SIGNAL	1	SPLICE-O	0.37	96	1,677	32,708	319,468	95,840	87,354
SIGNAL	2	BASE	0.44	10	2,844	84,801	553,834	166,150	191,534
SIGNAL	2	SPLICE-I	0.27	57	2,159	34,151	442,731	132,819	127,171
SIGNAL	2	SPLICE-O	0.33	57	2,092	28,693	319,468	95,840	87,354

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
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Arm Connection Analysis (Extreme I)

Analysis of Signal/Sign Arm Simplex Bolts

Mast Arm	Max Bolt CFI	Applied Forces (kip)		Factored Resistance (kip)	
		Tension	Shear	Tension	Shear
1	0.15	34.48	0.63	89.54	44.77
2	0.17	36.52	0.71	89.54	44.77

Analysis of Signal/Sign Arm Simplex Plates

Member Type	Max CSR	Applied Moment (ft-lb)	Factored Resistance (ft-lb)	Angle of Failure Line (deg)	Length of Bend Line (in)
1	0.15	17,692.33	117,779.92	45	19.39
2	0.16	18,739.91	117,779.92	45	19.39

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
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VERSION: 29.0.17.35

SUBJECT: JIMMY CARTER BLVD-POLE C-P30' M55'&M55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P30M5555A15115

Pole: Wind and Weight Force Data (Extreme I)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft2)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)
ATTCHMT. 1	20.0000	39.0000	8.67	1.20	0.90	0.85	29.44	306
ATTCHMT. 2	20.0000	45.0000	2.60	1.23	0.90	0.85	29.44	94
ATTCHMT. 3	20.0000	51.0000	8.67	1.20	0.90	0.85	29.44	306
ATTCHMT. 4	20.0000	24.0000	8.67	1.20	0.90	0.85	29.44	306
ATTCHMT. 5	20.0000	30.0000	9.74	1.20	0.90	0.85	29.44	344
ATTCHMT. 6	20.0000	36.0000	8.67	1.20	0.90	0.85	29.44	306
ATTCHMT. 7	20.0000	42.0000	7.49	1.19	0.90	0.85	29.44	262
ATTCHMT. 8	20.0000	48.0000	11.20	1.20	0.90	0.85	29.44	396
ATTCHMT. 9	30.0000	1.0000	2.00	1.00	0.98	0.85	32.07	64
30.0000	28.7439	0.0000	2.49	0.45	0.97	0.85	31.78	36
27.5000	26.2441	0.0000	2.57	0.45	0.95	0.85	31.18	36
25.0000	23.7442	0.0000	2.64	0.45	0.93	0.85	30.53	36
22.5000	21.2444	0.0000	2.71	0.45	0.91	0.85	29.82	36
20.0000	18.7445	0.0000	2.79	0.45	0.89	0.85	29.04	36
17.5000	16.2447	0.0000	2.86	0.45	0.86	0.85	28.18	36
15.0000	13.7448	0.0000	2.93	0.45	0.86	0.85	28.09	37
12.5000	11.2449	0.0000	3.01	0.45	0.86	0.85	28.09	38
10.0000	8.7451	0.0000	3.08	0.45	0.86	0.85	28.09	39
7.5000	6.2452	0.0000	3.15	0.45	0.86	0.85	28.09	40
5.0000	3.7453	0.0000	3.22	0.45	0.86	0.85	28.09	41
2.5000	1.2454	0.0000	3.30	0.45	0.86	0.85	28.09	42

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
IN ACCORDANCE WITH AASHTO-2015 RQMTS. (FINAL DEFLECTED POSITION)

BY: SE70

08/18/2026

VERSION: 29.0.17.35

SUBJECT: JIMMY CARTER BLVD-POLE C-P30' M55'&M55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P30M5555A15115

Pole: Forces and Moments (Extreme I)

Section Height* (ft)	Shear (lb)		Axial (lb)	Moment (ft-lb)		Torsion (ft-lb)	Wind Direct** (deg)
	Fx	Fy	Fz	Mx	My	Mz	
30.00	-63.07	-12.79	9.63	1.37	11.00	62.71	190.00
20.00	-3,139.89	-662.53	3,946.63	42,711.59	-37,900.21	-67,094.42	190.00
0.00	-3,391.70	-604.55	5,082.15	55,055.13	-103,440.88	-67,095.56	190.00

* These heights are above the pole base plate.

** These are directions toward which the wind is flowing.
They are angles from the +X axis in the X-Y plane.

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
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08/18/2026

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File: P30M5555A15115

Pole: Resistances (Extreme I)

Section Height* (ft)	Comb. Force Inter.	Applied Force				Factored Resistance			
		Axial (lb)	Shear (lb)	Bend. (ft-lb)	Torsion (ft-lb)	Axial $\phi=0.9$ (lb)	Shear $\phi=0.9$ (lb)	Bend. $\phi=0.9$ (ft-lb)	Torsion $\phi=0.95$ (ft-lb)
30.00	0.00	9.63	64.35	11.08	62.71	556,353.51	167,424.14	168,836.70	169,144.17
20.00	0.39	3,946.63	3,209.02	57,102.59	67,094.42	489,431.27	187,828.86	207,429.51	212,884.15
0.00	0.47	5,082.15	3,445.16	117,179.70	67,095.56	187,018.38	228,638.31	296,279.46	315,437.59

* These heights are above the pole base plate.

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Baseplate Analysis (Extreme I) - Pole1 - Pole

Combined Force Interaction	0.26
Critical Wind Direction *	220.00 deg
Alignment of Bend Line	135.00 deg
Width of Bending Section	16.527 (in)
Failure Line Start Coordinate (in)	(-11.500, 0.186)
Failure Line End Coordinate in	(0.186, -11.500)
Applied Bending Moment	18,217.33 ft-lb
Factored Bending Resistance	69,722.91 ft-lb
Plate Controlling Bolt Forces	

Bolt Number	Axial force (lb)	Moment Arm (in)	Bending moment (ft-lb)
3	-67264	3.25	-18217

Anchor Bolts Analysis (Extreme I) - Pole1 - Pole

Critical Wind Direct.* (deg)	Comb. Force Inter.	Applied Stress (psi)		Factored Resistance (psi)	
		Axial	Shear	ϕ F'nt	ϕ Fv
220.00	0.40	16,917.17	854.17	42,187.50	22,500.00

* Per AISC Design Guide 1

* These are directions toward which the wind is flowing

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Pole Deflection Information: (Extreme I)

Critical Wind Direction: 190.00

Height (ft)	X-Defl. (in)	Y-Defl. (in)	Deflection Angle - X (deg)	Deflection Angle - Y (deg)
30.0000	-4.74	-3.16	-1.09	0.79
27.5000	-4.17	-2.76	-1.09	0.79
25.0000	-3.59	-2.35	-1.09	0.79
22.5000	-3.01	-1.94	-1.09	0.79
20.0000	-2.44	-1.54	-1.09	0.79
17.5000	-1.89	-1.16	-0.98	0.67
15.0000	-1.41	-0.84	-0.86	0.56
12.5000	-0.99	-0.58	-0.73	0.46
10.0000	-0.64	-0.37	-0.59	0.36
7.5000	-0.36	-0.20	-0.45	0.26
5.0000	-0.16	-0.09	-0.31	0.17
2.5000	-0.04	-0.02	-0.15	0.08
0.0000	0.00	0.00	0.00	0.00

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SERVICE I LIMIT STATE

Wind Velocity	76.0 mph
Dead Component Load Factor	1.00
Wind Load Factor	1.00
Gust Factor	1.14

Mast Arm 1: Wind and Weight Force Data (Service I)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft2)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)	Notes
ATTCHMT. 1	20.0000	39.0000	8.67	1.20	0.90	0.85	12.86	134	
ATTCHMT. 2	20.0000	45.0000	15.00	1.23	0.90	0.85	12.86	237	
ATTCHMT. 3	20.0000	51.0000	8.67	1.20	0.90	0.85	12.86	134	

Mast Arm 2: Wind and Weight Force Data (Service I)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft2)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)	Notes
ATTCHMT. 4	20.0000	24.0000	8.67	1.20	0.90	0.85	12.86	134	
ATTCHMT. 5	20.0000	30.0000	9.74	1.20	0.90	0.85	12.86	150	
ATTCHMT. 6	20.0000	36.0000	8.67	1.20	0.90	0.85	12.86	134	
ATTCHMT. 7	20.0000	42.0000	7.50	1.19	0.90	0.85	12.86	115	
ATTCHMT. 8	20.0000	48.0000	11.20	1.20	0.90	0.85	12.86	173	

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
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Arm: Forces and Moments (Service I)

Arm Type	Arm No.	Analysis Location	Forces (lb)			Moment (ft-lb)		
			Axial	Fy	Fz	Torsion	My	Mz
SIGNAL	1	BASE	28.04	864.44	1,561.80	1.61	37,610.52	31,455.42
SIGNAL	1	SPLICE-I	33.83	685.60	845.12	0.17	13,298.08	15,498.79
SIGNAL	1	SPLICE-O	32.44	683.49	707.07	0.08	11,303.89	13,739.83
SIGNAL	2	BASE	2.46	1,028.00	1,627.12	1.33	38,969.42	33,491.77
SIGNAL	2	SPLICE-I	11.05	868.29	910.29	0.01	13,324.57	14,015.46
SIGNAL	2	SPLICE-O	9.29	866.27	772.27	0.01	11,162.92	11,787.01

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Arm: Resistances (Service I)

Analysis Location			Comb. Force Inter.	Applied Forces			Factored Resistance Forces		
Arm Type	Arm No.	Site		Axial (lb)	Shear (lb)	Bend. (ft-lb)	Axial $\phi=0.9$ (lb)	Shear $\phi=0.9$ (lb)	Bend. $\phi=0.9$ (ft-lb)
SIGNAL	1	BASE	0.26	28	1,785	49,031	553,834	166,150	191,534
SIGNAL	1	SPLICE-I	0.16	34	1,088	20,422	442,731	132,819	127,171
SIGNAL	1	SPLICE-O	0.20	32	983	17,792	319,468	95,840	87,354
SIGNAL	2	BASE	0.27	2	1,925	51,384	33,363	166,150	191,534
SIGNAL	2	SPLICE-I	0.15	11	1,258	19,338	442,731	132,819	127,171
SIGNAL	2	SPLICE-O	0.19	9	1,161	16,234	319,468	95,840	87,354

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Arm Connection Analysis (Service I)

Analysis of Signal/Sign Arm Simplex Bolts

Mast Arm	Max Bolt CFI	Applied Forces (kip)		Factored Resistance (kip)	
		Tension	Shear	Tension	Shear
1	0.06	21.82	0.45	89.54	44.77
2	0.07	22.88	0.48	89.54	44.77

Analysis of Signal/Sign Arm Simplex Plates

Member Type	Max CSR	Applied Moment (ft-lb)	Factored Resistance (ft-lb)	Angle of Failure Line (deg)	Length of Bend Line (in)
1	0.10	11,196.44	117,779.92	45	19.39
2	0.10	11,743.39	117,779.92	45	19.39

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Pole: Wind and Weight Force Data (Service I)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft2)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)
ATTCHMT. 1	20.0000	39.0000	8.67	1.20	0.90	0.85	12.86	134
ATTCHMT. 2	20.0000	45.0000	11.49	1.23	0.90	0.85	12.86	182
ATTCHMT. 3	20.0000	51.0000	8.67	1.20	0.90	0.85	12.86	134
ATTCHMT. 4	20.0000	24.0000	8.67	1.20	0.90	0.85	12.86	134
ATTCHMT. 5	20.0000	30.0000	7.13	1.20	0.90	0.85	12.86	110
ATTCHMT. 6	20.0000	36.0000	8.67	1.20	0.90	0.85	12.86	134
ATTCHMT. 7	20.0000	42.0000	5.49	1.19	0.90	0.85	12.86	84
ATTCHMT. 8	20.0000	48.0000	11.20	1.20	0.90	0.85	12.86	173
ATTCHMT. 9	30.0000	1.0000	2.00	1.00	0.98	0.85	14.01	28
30.0000	28.7439	0.0000	2.49	0.46	0.97	0.85	13.88	16
27.5000	26.2441	0.0000	2.57	0.45	0.95	0.85	13.62	16
25.0000	23.7442	0.0000	2.64	0.45	0.93	0.85	13.33	16
22.5000	21.2444	0.0000	2.71	0.45	0.91	0.85	13.02	16
20.0000	18.7445	0.0000	2.79	0.45	0.89	0.85	12.68	16
17.5000	16.2447	0.0000	2.86	0.45	0.86	0.85	12.31	16
15.0000	13.7448	0.0000	2.93	0.45	0.86	0.85	12.27	16
12.5000	11.2449	0.0000	3.01	0.45	0.86	0.85	12.27	17
10.0000	8.7451	0.0000	3.08	0.45	0.86	0.85	12.27	17
7.5000	6.2452	0.0000	3.15	0.45	0.86	0.85	12.27	17
5.0000	3.7453	0.0000	3.22	0.45	0.86	0.85	12.27	18
2.5000	1.2454	0.0000	3.30	0.45	0.86	0.85	12.27	18

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
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BY: SE70

08/18/2026

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SUBJECT: JIMMY CARTER BLVD-POLE C-P30' M55'&M55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P30M5555A15115

Pole: Forces and Moments (Service I)

Section Height* (ft)	Shear (lb)		Axial (lb)	Moment (ft-lb)		Torsion (ft-lb)	Wind Direct** (deg)
	Fx	Fy	Fz	Mx	My	Mz	
30.00	-18.11	-21.60	9.47	0.52	10.00	17.85	230.00
20.00	-946.95	-1,128.87	3,619.14	39,858.68	-33,906.67	-1,157.40	230.00
0.00	-996.41	-1,187.49	4,619.55	63,017.85	-53,343.46	-1,157.39	230.00

* These heights are above the pole base plate.

** These are directions toward which the wind is flowing.
They are angles from the +X axis in the X-Y plane.

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
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Folder: GAGWIN

File: P30M5555A15115

Pole: Resistances (Service I)

Section Height* (ft)	Comb. Force Inter.	Applied Force				Factored Resistance			
		Axial (lb)	Shear (lb)	Bend. (ft-lb)	Torsion (ft-lb)	Axial $\phi=0.9$ (lb)	Shear $\phi=0.9$ (lb)	Bend. $\phi=0.9$ (ft-lb)	Torsion $\phi=0.95$ (ft-lb)
30.00	0.00	9.47	28.19	10.01	17.85	556,353.51	167,424.14	168,836.70	169,144.17
20.00	0.26	3,619.14	1,473.45	52,329.50	1,157.40	489,431.27	187,828.86	207,429.51	212,884.15
0.00	0.29	4,619.55	1,550.15	82,563.75	1,157.39	187,018.38	228,638.31	296,279.46	315,437.59

* These heights are above the pole base plate.

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
IN ACCORDANCE WITH AASHTO-2015 RQMTS. (FINAL DEFLECTED POSITION)

BY: SE70

08/18/2026

VERSION: 29.0.17.35

SUBJECT: JIMMY CARTER BLVD-POLE C-P30' M55'&M55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P30M5555A15115

Baseplate Analysis (Service I) - Pole1 - Pole

Combined Force Interaction	0.18
Critical Wind Direction *	220.00 deg
Alignment of Bend Line	135.00 deg
Width of Bending Section	16.527 (in)
Failure Line Start Coordinate (in)	(-11.500, 0.186)
Failure Line End Coordinate in	(0.186, -11.500)
Applied Bending Moment	12,219.04 ft-lb
Factored Bending Resistance	69,722.91 ft-lb
Plate Controlling Bolt Forces	

Bolt Number	Axial force (lb)	Moment Arm (in)	Bending moment (ft-lb)
3	-45116	3.25	-12219

Anchor Bolts Analysis (Service I) - Pole1 - Pole

Critical Wind Direct.* (deg)	Comb. Force Inter.	Applied Stress (psi)		Factored Resistance (psi)	
		Axial	Shear	$\phi F'_{nt}$	ϕF_v
220.00	0.27	11,346.97	386.59	42,187.50	22,500.00

* Per AISC Design Guide 1

* These are directions toward which the wind is flowing

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File: P30M5555A15115

Pole Deflection Information: (Service I)

Critical Wind Direction: 230.00

Height (ft)	X-Defl. (in)	Y-Defl. (in)	Deflection Angle - X (deg)	Deflection Angle - Y (deg)
30.0000	-2.87	-3.38	-0.69	0.82
27.5000	-2.50	-2.95	-0.69	0.82
25.0000	-2.14	-2.52	-0.69	0.82
22.5000	-1.78	-2.10	-0.69	0.82
20.0000	-1.42	-1.67	-0.69	0.81
17.5000	-1.08	-1.27	-0.60	0.71
15.0000	-0.79	-0.93	-0.51	0.60
12.5000	-0.54	-0.64	-0.42	0.50
10.0000	-0.35	-0.41	-0.33	0.39
7.5000	-0.19	-0.23	-0.25	0.29
5.0000	-0.09	-0.10	-0.16	0.19
2.5000	-0.02	-0.03	-0.08	0.10
0.0000	0.00	0.00	0.00	0.00

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BY: SE70
08/18/2026
SUBJECT: JIMMY CARTER BLVD-POLE C-P30' M55'&M55' AASHTO 2015 115MPH
Folder: GAGWIN
ICE LIMIT STATE

VERSION: 29.0.17.35

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Wind Velocity	76.0 mph
Dead Component Load Factor	1.10
Wind Load Factor	1.00
Gust Factor	1.14

Mast Arm 1: Wind and Weight Force Data (Ice)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft2)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)	Notes
ATTCHMT. 1	20.0000	39.0000	8.67	1.20	0.90	0.85	12.86	134	
ATTCHMT. 2	20.0000	45.0000	15.00	1.23	0.90	0.85	12.86	237	
ATTCHMT. 3	20.0000	51.0000	8.67	1.20	0.90	0.85	12.86	134	

Mast Arm 2: Wind and Weight Force Data (Ice)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft2)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)	Notes
ATTCHMT. 4	20.0000	24.0000	8.67	1.20	0.90	0.85	12.86	134	
ATTCHMT. 5	20.0000	30.0000	9.74	1.20	0.90	0.85	12.86	150	
ATTCHMT. 6	20.0000	36.0000	8.67	1.20	0.90	0.85	12.86	134	
ATTCHMT. 7	20.0000	42.0000	7.50	1.19	0.90	0.85	12.86	115	
ATTCHMT. 8	20.0000	48.0000	11.20	1.20	0.90	0.85	12.86	173	

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Folder: GAGWIN

File: P30M5555A15115

Arm: Forces and Moments (Ice)

Arm Type	Arm No.	Analysis Location	Forces (lb)			Moment (ft-lb)		
			Axial	Fy	Fz	Torsion	My	Mz
SIGNAL	1	BASE	64.86	905.80	2,586.89	2.04	69,377.23	32,605.92
SIGNAL	1	SPLICE-I	81.15	711.48	1,567.06	0.24	27,338.06	15,967.82
SIGNAL	1	SPLICE-O	76.98	706.41	1,370.63	0.11	23,563.70	14,146.09
SIGNAL	2	BASE	44.61	1,070.89	2,850.92	1.62	75,260.35	34,657.35
SIGNAL	2	SPLICE-I	73.55	897.05	1,830.44	0.01	27,836.04	14,453.85
SIGNAL	2	SPLICE-O	68.30	892.36	1,634.08	0.01	23,384.91	12,154.87

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Folder: GAGWIN

File: P30M5555A15115

Arm: Resistances (Ice)

Analysis Location			Comb. Force Inter.	Applied Forces			Factored Resistance Forces		
Arm Type	Arm No.	Site		Axial (lb)	Shear (lb)	Bend. (ft-lb)	Axial $\phi=0.9$ (lb)	Shear $\phi=0.9$ (lb)	Bend. $\phi=0.9$ (ft-lb)
SIGNAL	1	BASE	0.40	65	2,741	76,657	553,834	166,150	191,534
SIGNAL	1	SPLICE-I	0.25	81	1,721	31,660	442,731	132,819	127,171
SIGNAL	1	SPLICE-O	0.31	77	1,542	27,484	319,468	95,840	87,354
SIGNAL	2	BASE	0.43	45	3,045	82,857	553,834	166,150	191,534
SIGNAL	2	SPLICE-I	0.25	74	2,038	31,365	442,731	132,819	127,171
SIGNAL	2	SPLICE-O	0.30	68	1,862	26,355	319,468	95,840	87,354

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Folder: GAGWIN

File: P30M5555A15115

Arm Connection Analysis (Ice)

Analysis of Signal/Sign Arm Simplex Bolts

Mast Arm	Max Bolt CFI	Applied Forces (kip)		Factored Resistance (kip)	
		Tension	Shear	Tension	Shear
1	0.13	32.22	0.69	89.54	44.77
2	0.15	34.72	0.76	89.54	44.77

Analysis of Signal/Sign Arm Simplex Plates

Member Type	Max CSR	Applied Moment (ft-lb)	Factored Resistance (ft-lb)	Angle of Failure Line (deg)	Length of Bend Line (in)
1	0.14	16,535.73	117,779.92	45	19.39
2	0.15	17,819.00	117,779.92	45	19.39

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BY: SE70

08/18/2026

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SUBJECT: JIMMY CARTER BLVD-POLE C-P30' M55'&M55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P30M5555A15115

Pole: Wind and Weight Force Data (Ice)

Elevation at Top of Section (ft)	Centroid Above Base (ft)	Ecc. From Pole Centerline (ft)	Section Projected Area (ft2)	Section Drag Coeff.	Kz	Kd	Wind Pressure (psf)	Wind Force (lb)
ATTCHMT. 1	20.0000	39.0000	8.67	1.20	0.90	0.85	12.86	134
ATTCHMT. 2	20.0000	45.0000	0.00	1.23	0.90	0.85	12.86	0
ATTCHMT. 3	20.0000	51.0000	8.67	1.20	0.90	0.85	12.86	134
ATTCHMT. 4	20.0000	24.0000	8.67	1.20	0.90	0.85	12.86	134
ATTCHMT. 5	20.0000	30.0000	9.68	1.20	0.90	0.85	12.86	149
ATTCHMT. 6	20.0000	36.0000	8.67	1.20	0.90	0.85	12.86	134
ATTCHMT. 7	20.0000	42.0000	7.44	1.19	0.90	0.85	12.86	114
ATTCHMT. 8	20.0000	48.0000	11.20	1.20	0.90	0.85	12.86	173
ATTCHMT. 9	30.0000	1.0000	2.00	1.00	0.98	0.85	14.01	28
30.0000	28.7439	0.0000	2.49	0.46	0.97	0.85	13.88	16
27.5000	26.2441	0.0000	2.57	0.45	0.95	0.85	13.62	16
25.0000	23.7442	0.0000	2.64	0.45	0.93	0.85	13.33	16
22.5000	21.2444	0.0000	2.71	0.45	0.91	0.85	13.02	16
20.0000	18.7445	0.0000	2.79	0.45	0.89	0.85	12.68	16
17.5000	16.2447	0.0000	2.86	0.45	0.86	0.85	12.31	16
15.0000	13.7448	0.0000	2.93	0.45	0.86	0.85	12.27	16
12.5000	11.2449	0.0000	3.01	0.45	0.86	0.85	12.27	17
10.0000	8.7451	0.0000	3.08	0.45	0.86	0.85	12.27	17
7.5000	6.2452	0.0000	3.15	0.45	0.86	0.85	12.27	17
5.0000	3.7453	0.0000	3.22	0.45	0.86	0.85	12.27	18
2.5000	1.2454	0.0000	3.30	0.45	0.86	0.85	12.27	18

ANALYSIS OF VALMONT INDUSTRIES LIGHTING STRUCTURE
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BY: SE70

08/18/2026

VERSION: 29.0.17.35

SUBJECT: JIMMY CARTER BLVD-POLE C-P30' M55'&M55' AASHTO 2015 115MPH

Folder: GAGWIN

File: P30M5555A15115

Pole: Forces and Moments (Ice)

Section Height* (ft)	Shear (lb)		Axial (lb)	Moment (ft-lb)		Torsion (ft-lb)	Wind Direct** (deg)
	Fx	Fy	Fz	Mx	My	Mz	
30.00	-28.80	-1.14	36.78	0.60	37.38	27.18	180.00
20.00	-1,446.40	-137.09	6,061.23	76,465.80	-62,329.91	-32,428.85	180.00
0.00	-1,469.94	-8.58	7,415.56	77,738.77	-91,661.87	-32,429.95	180.00

* These heights are above the pole base plate.

** These are directions toward which the wind is flowing.
They are angles from the +X axis in the X-Y plane.

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Pole: Resistances (Ice)

Section Height* (ft)	Comb. Force Inter.	Applied Force				Factored Resistance			
		Axial (lb)	Shear (lb)	Bend. (ft-lb)	Torsion (ft-lb)	Axial $\phi=0.9$ (lb)	Shear $\phi=0.9$ (lb)	Bend. $\phi=0.9$ (ft-lb)	Torsion $\phi=0.95$ (ft-lb)
30.00	0.00	36.78	28.82	37.39	27.18	556,353.51	167,424.14	168,836.70	169,144.17
20.00	0.48	6,061.23	1,452.89	98,651.08	32,428.85	489,431.27	187,828.86	207,429.51	212,884.15
0.00	0.43	7,415.56	1,469.97	120,188.24	32,429.95	187,018.38	228,638.31	296,279.46	315,437.59

* These heights are above the pole base plate.

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Folder: GAGWIN

File: P30M5555A15115

Baseplate Analysis (Ice) - Pole1 - Pole

Combined Force Interaction	0.27
Critical Wind Direction *	220.00 deg
Alignment of Bend Line	135.00 deg
Width of Bending Section	16.527 (in)
Failure Line Start Coordinate (in)	(-11.500, 0.186)
Failure Line End Coordinate in	(0.186, -11.500)
Applied Bending Moment	19,143.26 ft-lb
Factored Bending Resistance	69,722.91 ft-lb
Plate Controlling Bolt Forces	

Bolt Number	Axial force (lb)	Moment Arm (in)	Bending moment (ft-lb)
3	-70683	3.25	-19143

Anchor Bolts Analysis (Ice) - Pole1 - Pole

Critical Wind Direct.* (deg)	Comb. Force Inter.	Applied Stress (psi)		Factored Resistance (psi)	
		Axial	Shear	ϕ F'nt	ϕ Fv
220.00	0.42	17,777.02	387.46	42,187.50	22,500.00

* Per AISC Design Guide 1

* These are directions toward which the wind is flowing

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Pole Deflection Information: (Ice)

Critical Wind Direction: 180.00

Height (ft)	X-Defl. (in)	Y-Defl. (in)	Deflection Angle - X (deg)	Deflection Angle - Y (deg)
30.0000	-5.08	-4.98	-1.23	1.25
27.5000	-4.43	-4.33	-1.23	1.25
25.0000	-3.78	-3.68	-1.23	1.25
22.5000	-3.14	-3.02	-1.23	1.25
20.0000	-2.49	-2.37	-1.23	1.25
17.5000	-1.89	-1.77	-1.06	1.05
15.0000	-1.38	-1.27	-0.90	0.87
12.5000	-0.95	-0.86	-0.74	0.70
10.0000	-0.60	-0.54	-0.58	0.54
7.5000	-0.34	-0.30	-0.43	0.39
5.0000	-0.15	-0.13	-0.28	0.25
2.5000	-0.04	-0.03	-0.14	0.12
0.0000	0.00	0.00	0.00	0.00

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FATIGUE II LIMIT STATE

Galloping	No
Natural Wind Gust (11.2 mph)	Yes
Truck-Induced Gust (65.0 mph)	No
Importance Factor	II

Mast Arm: Fatigue Analysis (Fatigue II)

Analysis Location			Design Load	Comb. Force Inter.	Moment (ft-lb)	Shear force (lb)	Shear Stress (ksi)	Applied Bending Stress (ksi)	Allowable Stress (ksi)
Arm Type	Arm No.	Site							
MA	1	BASE	NATURAL WIND GUST	0.86	12,888.85	390	0.07	3.88	4.50
MA	1	SP-I	NATURAL WIND GUST	0.24	5,979.31	277	0.06	2.82	12.00
MA	1	SP-O	NATURAL WIND GUST	0.29	5,266.48	277	0.09	3.42	12.00
MA	2	BASE	NATURAL WIND GUST	0.89	13,221.45	430	0.08	3.98	4.50
MA	2	SP-I	NATURAL WIND GUST	0.21	5,384.72	330	0.07	2.54	12.00
MA	2	SP-O	NATURAL WIND GUST	0.25	4,537.78	330	0.10	2.95	12.00

Mast Arm: Deflections (Fatigue II)

Arm Type	Arm No.	Load Case	Max Vertical Deflection (in)
MA	1	NATURAL WIND GUST	1.11
MA	2	NATURAL WIND GUST	1.14

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Fatigue Analysis of Signal and Sign / Pole Connection: Arm

Arm Type	Arm No.	Component	Load	Stress Ratio	Applied Stress (ksi)	Allowable Stress (ksi)
MA	1	SIMPLEX BOLT	NATURAL WIND GUST	0.60	4.21	7.00
MA	2	SIMPLEX BOLT	NATURAL WIND GUST	0.62	4.32	7.00

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Analysis of Pole (Fatigue II)

Section Height* (ft)	Design Load	Comb. Force Inter.	Moment (ft-lb)	Applied Bending Stress (ksi)	Allowable Bending Stress (ksi)	Deflection (in)
0.00	NATURAL WIND GUST	0.58	13,097.60	2.61	4.50	0.00

* These heights are above the pole base plate.

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Tube to Transverse Plate CAFT Calculation Details

	Pole to Baseplate Weld	Arm 55M Shaft to Simplex Plate Weld
Weld type	Socket	Socket
Tt (in)	0.31250	0.25000
Dt (in)	16.00	14.50
Ttp (in)	2.50	3.00
Dbc (in)	22.50	26.87
CBC	1.41	1.85
Dop (in)	N/A	N/A
COP	N/A	N/A
NS	0.00	0.00
RRb (in)	N/A	N/A
Multisided Factor	N/A	N/A
Kf	2.77	2.67
Ki	6.26	5.69
CAFT (ksi)	4.50	4.50

NOTE: The maximum bolt circle is used for bolt patterns where all the bolts do not lie on a single circle, per AASHTO.

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Fatigue Analysis of Anchor Bolts (NATURAL WIND GUST)

Load Case	Combined Stress Ratio	Axial (lb)
NATURAL WIND GUST	0.30	6,772.25

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Opening Group on the Pole

Description	Attachment Height (ft)	Clear Opening (in)			Reinforcement (in)		
		Width	Height	Inside Corner Radius	Rim Thickness	Rim Depth	Rim Projection
STD HH	2.75	4.48	7.00	2.24	0.28	2.50	0.50

Description	Location On Pole (ft)	Orientation (deg)	Tube Diam. (in)	Tube Thick. (in)	Area (in ²)	X Centroid (in)	Y Centroid (in)	Ix(in ⁴)	Iy(in ⁴)
STD HH	2.75	0.00	15.62	0.31	14.89	-0.14	0.00	448	424

		Stress at Root		Stress at Toe		
Description	Moment (ft-lb)	Actual (ksi)	Resist (ksi)	Actual (ksi)	Resist (ksi)	Max CSR
NATURAL WIND GUST	11,318.87	10.10	16.00	2.52	7.00	0.63